

# Observable Gauge Algebra from Finite-Rank Majorana Projection

Pasquale Camelia\*

June 2026

## Abstract

In a closed system with no external clock — the situation a Wheeler–DeWitt constraint describes — a finite observer never reads the full state, only a projection of it. We ask which gauge structure survives that projection. Global conservation under such restricted access forces a self-conjugate (Majorana) carrier and a finite-rank, non-invertible projection onto the boundary; the observables are the projected bilinears, and the gauge algebra is their stabiliser on the quotient by the projection kernel. We show that this stabiliser is a compact  $\mathfrak{su}(2)$  algebra with a distinguished Abelian Cartan direction, fixed by the metric degeneracy of the boundary and the Majorana symplectic structure, and reproduced independently by the braid representation carried by the finite-rank truncation. From the dimensions of these sectors a single internal ratio follows; each of its ingredients is fixed by the structure of the projection rather than by any chosen constant. Where the structure stops forcing — the selection of the exchange sector, the identification of the braid algebra with the boundary gauge algebra, and the matching of the internal ratio to the electroweak mixing angle — the paper states the step as open rather than closed. The gauge-algebra results are purely algebraic and use no field equation. An appendix develops, as an independent dynamical compatibility layer resting on a boundary Dirac operator, the conformal slab and the Cayley–Hardy kernel quadrature; the algebraic results do not depend on it. Masses, running couplings, Yukawa matrices, and QCD dynamics lie outside the present scope.

**Keywords.** Wheeler–DeWitt constraint; non-invertible projection; Majorana spinor; boundary gauge algebra; Temperley–Lieb algebra; electroweak mixing; problem of time; relational observables.

## 1 Introduction

A closed quantum system has no external observer and no external clock, yet a boundary still reads a definite gauge content from it. This paper asks what that content is forced to be when the only access is through a non-invertible projection, and shows that within the imported rank-five Majorana admissibility class, the projection fixes the observable stabiliser algebra and an internal dimension ratio — the fraction of the boundary internal dimension carried by the non-Abelian adjoint channels. Each ingredient of that ratio (the rank, the chiral doubling, the adjoint channel count) is fixed by the structure of the projection; its numerical value is a readout, recorded once the structure is in place. We begin with the readout problem in its standard canonical setting and then make the limited access explicit as an operator.

---

\*Independent Researcher, Milan, Italy. [pasquale.camelia@gmail.com](mailto:pasquale.camelia@gmail.com). ORCID: [0009-0006-4779-4647](https://orcid.org/0009-0006-4779-4647). © 2026 Pasquale Camelia. Licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/).

## 1.1 Constrained closed systems and the readout problem

In canonical quantum gravity the Wheeler–DeWitt constraint [1, 2] is the standard expression of a closed, background-independent system with no external clock, written as a condition on the physical Hilbert space:

$$\hat{\mathcal{H}}_{\text{bulk}} \Xi = 0. \quad (1)$$

A state satisfying (1) has no preferred external time: the Hamiltonian acts as a constraint on physical states, not as a generator of evolution against any background clock [9, 11]. We do not assume that every fundamental theory must literally be of Wheeler–DeWitt form. We take this constraint as the canonical representative of a broader physical situation — a closed constrained bulk accessible only through internal, finite readouts — and ask a question that is conditional and structural rather than ontological: given such a bulk carrier and a finite non-invertible boundary projection, what is the algebra of transformations that preserves the projected observables?

There is an associated readout question, often hidden inside the problem of time and less frequently made explicit. A finite boundary cannot read the whole constrained state; it reads some projection of it. Which observables does that projection leave accessible, and which transformations leave them invariant? We call this the readout problem, and it accompanies the problem of time rather than replacing it.

The question is not new. Relational quantum mechanics [3, 4] holds that facts are relative to the observing system; decoherent histories [5, 6] drop the measurement postulate in favour of consistent coarse-grained records; quantum Darwinism [7, 8] ties the emergence of classical observables to redundant encoding in the environment. What these share is the recognition that the observer has limited access, and that what counts as observable depends on the channel that remains open.

Here that limited access is made explicit and given a definite operator. The boundary receives only what a non-invertible projection  $\Pi$  lets through, and the kernel of  $\Pi$  is not the observer’s ignorance but a structured part of the carrier — the memory of what cannot be read:

$$\Psi_{\text{bdy}} = \Pi \Xi, \quad \ker \Pi \neq 0, \quad \Xi = \Xi_{\text{vis}} + \Xi_{\text{ker}}, \quad \Xi_{\text{ker}} \in \ker \Pi. \quad (2)$$

Here  $\Xi_{\text{vis}}$  is the component carried to the boundary and  $\Xi_{\text{ker}} = P_{\text{ker}} \Xi$  its kernel complement, with  $P_{\text{ker}}$  the projector onto  $\ker \Pi$ ; since  $\Pi$  maps into  $\mathcal{H}_{\text{obs}}$  rather than acting as an idempotent on  $\mathcal{H}_{\text{bulk}}$ , the split is written through the separate projector  $P_{\text{ker}}$  rather than as  $(1 - \Pi)$ .

## 1.2 Gauge symmetry from the projection

Every physical observable available to the boundary is a projected Majorana bilinear:

$$\bar{\Xi} \Gamma \Xi \xrightarrow{\Pi} \Psi_{\text{bdy}}^- \Gamma \Psi_{\text{bdy}}. \quad (3)$$

Two bulk configurations  $\Xi$  and  $\Xi' = \Xi + \chi$  with  $\chi \in \ker \Pi$  produce identical boundary bilinears:  $\Pi \chi = 0$  implies  $\Psi_{\text{bdy}}^- \Gamma \Psi_{\text{bdy}}$  is unchanged. To state this precisely, let  $\mathcal{B}_{\partial}$  be the *observable algebra*: the finite set of boundary bilinear forms  $G_{\Gamma}$  on  $\mathcal{H}_{\text{obs}} = \mathcal{H}_{\text{bulk}} / \ker \Pi$ ,  $G_{\Gamma}(u, v) = \langle u, \Gamma v \rangle_{K_{\Pi}}$ , generated by the admissible  $\Gamma$  in the projected Clifford module and the  $K_{\Pi}$ -adjoint. A transformation  $X$  on  $\mathcal{H}_{\text{bulk}}$  descends to  $\mathcal{H}_{\text{obs}}$  when  $X(\ker \Pi) \subseteq \ker \Pi$ , giving  $X_{\partial} \in \text{End}(\mathcal{H}_{\text{obs}})$  with  $\Pi X = X_{\partial} \Pi$ . The gauge algebra is the set of descended transformations that leave *every* form in the observable algebra invariant, as forms (for all states):

$$\mathfrak{g}_{\partial} = \{X_{\partial} \in \text{End}(\mathcal{H}_{\text{obs}}) \mid X_{\partial}^{\dagger} G_{\Gamma} + G_{\Gamma} X_{\partial} = 0 \ \forall G_{\Gamma} \in \mathcal{B}_{\partial}\}, \quad (4)$$

where  $^{\dagger}$  is the  $K_{\Pi}$ -adjoint. This is the Lie algebra of the isometry group of the form system  $\mathcal{B}_{\partial}$ , hence a genuine Lie algebra: if  $X_{\partial}, Y_{\partial}$  each satisfy (4), then for every  $G_{\Gamma}$ ,

$$\begin{aligned} [X_{\partial}, Y_{\partial}]^{\dagger} G_{\Gamma} + G_{\Gamma} [X_{\partial}, Y_{\partial}] &= Y_{\partial}^{\dagger} X_{\partial}^{\dagger} G_{\Gamma} - X_{\partial}^{\dagger} Y_{\partial}^{\dagger} G_{\Gamma} + G_{\Gamma} X_{\partial} Y_{\partial} - G_{\Gamma} Y_{\partial} X_{\partial} \\ &= G_{\Gamma} Y_{\partial} X_{\partial} - G_{\Gamma} X_{\partial} Y_{\partial} + G_{\Gamma} X_{\partial} Y_{\partial} - G_{\Gamma} Y_{\partial} X_{\partial} = 0, \end{aligned}$$

using  $X_\partial^\dagger G_\Gamma = -G_\Gamma X_\partial$  and  $Y_\partial^\dagger G_\Gamma = -G_\Gamma Y_\partial$  (both from (4)). Therefore  $[X_\partial, Y_\partial] \in \mathfrak{g}_\partial$ , and the set is closed under the commutator. The condition  $X(\ker \Pi) \subseteq \ker \Pi$  ensures that  $X$  descends to a well-defined operator  $X_\partial$  on the quotient  $\mathcal{H}_{\text{bulk}}/\ker \Pi$ , equivalently  $\Pi X = X_\partial \Pi$ ; without it  $X$  could map a kernel vector into the observable image and the action on the quotient would not be defined. The kernel  $\ker \Pi$  supplies the null directions along which the bilinear is insensitive;  $\mathfrak{g}_\partial$  is the algebra of those quotient transformations that preserve the complete system of observable boundary forms.

Read this way, gauge symmetry is not a property of the bulk imposed from outside. It is the symmetry of what stays legible once the projection has thrown part of the carrier into the kernel. The boundary never sees  $\Xi$ ; it sees the bilinears, and the gauge algebra is the redundancy of that reading — the transformations that move  $\Xi$  without moving anything the boundary can record.

This turns the readout question into a determinate one. Everything the boundary can know is encoded in  $\mathfrak{g}_\partial$ , and  $\mathfrak{g}_\partial$  in turn is fixed by the rank of the projection, the self-conjugacy of the carrier, the admissible mode catalogue, the saturation/confinement modes of Section 4, and the observable channel count of Section 9. The rest of the paper computes  $\mathfrak{g}_\partial$  from these inputs.

### 1.3 Inputs

This paper builds on the projection-SVD construction of [21], to which the reader is referred for the constructions imported below; further references support specific points. We summarise what is imported and why it is needed, so that the argument here can be followed without reconstructing those proofs.

The starting point is a single physical requirement: that a globally conserved current  $\partial_I J^I = 0$  remain meaningful when the state is read only through restricted access. This requirement has two consequences, established in [18, 19]: the readout map must be a non-invertible projection  $\Pi$  (a finite boundary cannot carry the full current), and the carrier must be self-conjugate,  $\Xi = C\bar{\Xi}$  (Majorana), for the bilinear charges to be real. These two facts — a projection and a real carrier — are the foundational physical input; the further structural ingredients used below (the rank-five projection, the doublet degeneracy, the admissible catalogue, the hypotheses (H1)–(H3), and the anisotropy tensor) are imported as stated, not re-derived here.

Reference [21] supplies the concrete shape of that projection. Its singular-value decomposition gives a rank-five Hilbert-like boundary filter

$$H(z) = \frac{P_4(z)}{1 - \alpha z}, \quad P_4(z) = \sum_{k=0}^4 \sigma_k z^k, \quad \sigma_k \propto \varphi^{-k}, \quad (5)$$

whose single pole at  $z = \alpha^{-1}$  sets the projection threshold, and a boundary metric  $K_\Pi = P_5^\dagger P_5$  fixing the inner product the projected observables inherit. The threshold is itself structural, not fitted. It is the inverse square of a projection scale built from the Pincherle dilation operator  $\mathcal{D} = x D_x$  acting on the  $\varphi$ -autosimilar hierarchy: its increment at the quadratic level is  $\mathcal{D}(x^2)|_{x=\varphi} = 2\varphi^2$ , and the Fibonacci curvature is itself a function of  $\varphi$ ,

$$\kappa = \sqrt{5} = 2\left(\varphi - \frac{1}{2}\right), \quad (6)$$

so that their product is built from  $\varphi$  alone:

$$\Lambda = \mathcal{D}(x^2)|_{x=\varphi} \cdot \kappa = 2\varphi^2 \cdot 2\left(\varphi - \frac{1}{2}\right) = 4\varphi^2\left(\varphi - \frac{1}{2}\right) = 2\varphi^2\sqrt{5}. \quad (7)$$

The form  $4\varphi^2(\varphi - \frac{1}{2})$  exhibits  $\Lambda$  as a pure function of  $\varphi$ , with no coefficient inserted from outside. Applying the golden identity  $\varphi^2 = \varphi + 1$  then writes  $\Lambda$  in the  $\mathbb{Q}(\sqrt{5})$ -coordinates intrinsic to the golden field, which are consecutive Fibonacci numbers:

$$\Lambda = F_5 + F_4\sqrt{5} \quad (= 5 + 3\sqrt{5}), \quad (8)$$

the coordinates  $F_5, F_4$  being an output of the dilation structure, not chosen coefficients; the integers 5, 3 are only their numerical readout. The projection threshold is then

$$\alpha = \Lambda^{-2} = [4\varphi^2(\varphi - \tfrac{1}{2})]^{-2}, \quad (9)$$

and its reciprocal  $\alpha^{-1} = \Lambda^2$  is exactly the pole  $z = \alpha^{-1}$  of the Hilbert-like filter (5): the Pincherle projection scale and the filter pole are the same structural object in two representations — one as the dilation increment times the curvature, the other as the denominator root of  $H(z)$ . The numerical value of this pole is a readout, developed in Section 5; it plays no role in the gauge-algebra construction below. We take the operators as given; the present paper asks what gauge algebra they imply, not how they are derived.

Two further inputs are used for specific steps: the uniqueness of rank five within the Majorana–Fibonacci admissibility class [20], and the model-independent kernel structure [22]. The spinorial lift of the confined boundary mode into the kernel is a separate problem, outside the scope of the present paper.

The role of the Wheeler–DeWitt condition here is motivational, not axiomatic. The object actually studied below is the boundary stabiliser algebra of projected Majorana bilinears under a finite non-invertible projection. The result should therefore be read as a conditional boundary-algebra theorem: if a closed constrained carrier is accessed through a rank-five Majorana projection, the observable gauge grammar derived below follows. The Wheeler–DeWitt equation motivates the setting; the projection and the stabiliser algebra carry the result. The gauge-algebra result — the stabiliser algebra, the residue, and the dimension ratio — is algebraic: it concerns the stabiliser algebra of projected Majorana bilinears under a finite non-invertible projection, and uses no boundary Dirac equation. The conformal slab and kernel quadrature of Section A do invoke a boundary Dirac operator, but they form a separate dynamical layer; the algebraic results of Sections 1–11 neither assume nor depend on it.

One point of reading deserves a word, since the admissible modes are labelled by windings on a compact internal cycle. These modes are not closed time curves, and the cycle  $S_\chi^1(\tau)$  is an internal phasorial fibre of the carrier, not a periodic time direction. The  $n$ -values are holonomy classes of the projected boundary carrier — admissible internal loops of the projection — and nothing in the gauge-algebra result requires interpreting them temporally. Observable time, in the present framework, is the ordered and irreversible sequence of boundary readouts induced by the non-invertible projection; its thermodynamic and relational interpretation is a separate matter, developed in the companion work on emergent physical time and not used here.

## 2 Transport symmetries of the Majorana carrier

Before the projection selects what the boundary can read, the carrier itself has a definite transport structure: a self-conjugate Majorana field on a pre-geometric domain with a compact chiral fibre. The symmetry of that transport — a compact non-Abelian symmetry with a distinguished Abelian Cartan direction — is what later survives projection as the boundary gauge algebra. We first describe the domain, then extract this symmetry and the channels it acts on.

### 2.1 The pre-geometric domain

Before the projection acts, the Majorana carrier  $\Xi = C\bar{\Xi}$  propagates on a pre-geometric domain  $\mathcal{M}_{\text{pre}}^5 = X_{\text{bdy}} \times S_\chi^1(\tau)$ , where  $X_{\text{bdy}}$  is the four-dimensional boundary locus and  $S_\chi^1(\tau)$  is a compact internal chiral phase fibre [19]. The four-dimensional locus  $X_{\text{bdy}}$  is not assumed to be Minkowski spacetime; it becomes  $\mathbb{R}_{\text{eff}}^4$  only after coherent global gluing of boundary cells [19].

## 2.2 The phase symmetry $U(1)_\eta$

The Majorana carrier on the compact fibre  $S_\chi^1(\tau)$  satisfies the spinorial closure condition [19]

$$\Psi(\chi + 2\pi) = -\Psi(\chi). \quad (10)$$

Traversing the compact cycle once returns the carrier with a global sign reversal. If  $\lambda$  is an eigenvalue of the transport operator for one traversal, the inverse traversal acquires the spinorial sign and contributes  $-\lambda^{-1}$ . The conjugate pair is therefore  $(\lambda, -\lambda^{-1})$ , and under restricted accessibility the phase distinguishing the two orientations is lost.

A single Majorana carrier  $\Xi = C\bar{\Xi}$  admits no global phase rotation  $\Xi \mapsto e^{i\theta}\Xi$ : the reality condition forces  $e^{2i\theta} = 1$ , leaving only  $\theta \in \{0, \pi\}$ . The Abelian symmetry is therefore not a phase of the spinor but a rotation of the real chiral plane spanned by the two channels  $(\xi, \eta)$ :

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} \mapsto \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix}, \quad \text{Spin}(2) \simeq U(1)_\eta, \quad (11)$$

generated by the single generator of  $\text{Spin}(2)$  acting on the real Majorana–Nambu doublet [19]. The relevant mode is the lowest antiperiodic mode of the fibre (the mode with half-integer Fourier index  $n = 1/2$ , forced by the antiperiodic condition  $\Xi(\chi + 2\pi) = -\Xi(\chi)$ ), and not a Fourier  $n = 0$  mode. On its own this  $U(1)_\eta$  is too weak: once the plane orientation is projected away only the modulus survives, and the reduced sector retains no non-trivial invariant. The Abelian channel does not exhaust the stabiliser of the projected doublet; the non-Abelian compact component is derived from the degenerate boundary metric and the Majorana symplectic form in Section 7. The proposition below records only which group is minimal once a non-Abelian partner is required; it is not the origin of that partner, which is the metric degeneracy of Section 7.

## 2.3 The orientation symmetry $SU(2)_\xi$

The Abelian rotation  $U(1)_\eta$  does not by itself survive projection as a non-trivial stabiliser. Its single generator  $J_\eta$  has one functionally independent Casimir, the squared modulus  $|\xi|^2 + |\eta|^2$ ; once the plane orientation is in the kernel, the only  $U(1)_\eta$ -invariant left on the reduced doublet is that modulus, a multiple of the identity, whose stabiliser is all of  $U(2)$  and so encodes no boundary gauge content. A non-trivial conserved structure on the projected doublet therefore requires an invariant that is not central, i.e. a stabiliser strictly between the scalars and  $U(2)$ . Such a stabiliser is non-Abelian: any maximal torus of  $U(2)$  fixes a decomposition of the doublet into two lines, which projection of the relative phase would render two independent channels rather than one conserved flow, so the carrier symmetry cannot be contained in a torus. This fixes the *character* of the second component — non-Abelian, acting irreducibly on the doublet — without yet fixing the group; the group is determined in Section 7 from the degeneracy of the boundary metric, where the symplectic reduction  $U(2) \rightarrow USp(2) \cong SU(2)$  leaves no freedom. The proposition below records the minimality any such component must satisfy.

**Proposition 2.1** (Minimal non-Abelian carrier symmetry).  *$SU(2) \simeq Sp(1)$  is the minimal compact connected non-Abelian group admitting a faithful pseudoreal two-dimensional representation. It is therefore the smallest such group that can act faithfully and spinorially on the chiral doublet  $S_\chi$ .*

*Proof.*  $SU(2)$  has rank one and dimension three, and its fundamental representation is the smallest pseudoreal representation, of complex dimension two — exactly the double-valued carrier required by the spinorial closure (10) [19]. Among compact connected groups: an Abelian group is excluded because it is not non-Abelian;  $SO(3)$  is excluded because it carries no faithful two-dimensional (pseudoreal spinorial) representation. Hence  $SU(2)$  is the minimal compact connected non-Abelian group with a faithful pseudoreal doublet. We claim only minimality in this precise sense; we do not claim that no larger group could act, and the argument does not require such a claim.  $\square$   $\square$

Prior to projection there is no factorisation between the compact phase and the  $SU(2)_\xi$  structure: the two are structurally entangled within the conserved carrier. Under restricted accessibility the continuous organisation reorganises into the coset structure  $SU(2)/U(1)$ , where  $U(1)$  is the stabiliser of the compact phase [19]. The chiral shell  $S_\chi^2 = SU(2)/U(1)$  is therefore not a geometric space but the internal symmetry structure encoding the conserved flow once continuity is no longer accessible. This  $SU(2)_\xi$  is not the electroweak  $SU(2)$  of the Standard Model: it is the minimal non-Abelian carrier symmetry of Proposition 2.1.

## 2.4 Projected boundary channels

The two structures derived above are not independent: the real phase rotation  $U(1)_\eta$  of Section 2.2 is the maximal torus (Cartan) of the orientation symmetry  $SU(2)_\xi$ , generated by  $-i\sigma_2$ , so

$$U(1)_\eta \subset SU(2)_\xi, \quad (12)$$

and the minimal continuous symmetry content selected under the stated Majorana and fibred assumptions, before the projection acts, is  $SU(2)_\xi$  with  $U(1)_\eta$  its distinguished Cartan. Proposition 2.1 establishes the minimality of this content; it does not exclude every larger group.

Projected onto the boundary, the carrier organises into a chiral doublet  $q = (\xi, \eta)^\top$ , with  $\xi$  the longitudinal channel of  $SU(2)_\xi$  and  $\eta$  the transverse channel of  $U(1)_\eta$ . The single conserved quantity of the carrier descends to the Noether invariant

$$K_\tau = \xi \eta = \alpha_{\text{QGT}}^{-1}, \quad (13)$$

distinct from the projection metric  $K_\Pi = P_5^\dagger P_5$  of §1 (the two are different objects sharing no relation beyond notation, which is why they carry different subscripts). The relation  $K_\tau = \alpha_{\text{QGT}}^{-1}$  is taken here as a structural input on the chiral doublet; it is not derived from the SVD metric alone [21]. The two channels are read independently:  $P_\eta$  selects the transverse  $U(1)_\eta$  stabiliser channel, and  $P_\xi$  selects the complementary longitudinal stabiliser channel. The transverse channel is the one whose stabiliser stays unbroken in the analysis below, and which carries the invariant Abelian channel; the longitudinal channel is its complement. (Whether this Abelian channel is the physical electromagnetic axis is part of the conditional bridge of Section 7.) These two readouts, and the invariant (13) relating them, constitute the minimal continuous symmetry content selected under the stated Majorana and fibred assumptions; they are all that the gauge-algebra analysis below requires.

## 3 The admissible boundary modes

Not every winding mode of the chiral fibre reaches the boundary: only those whose minimal spinorial orbit closes inside the rank-five window become observable. This section fixes that catalogue and assigns each admissible mode its physical role, providing the finite vocabulary the rest of the paper draws on.

### 3.1 The admissible catalogue

The projection has rank  $r = 5$  [20], so a winding mode  $n$  on  $S_\chi^1$  becomes a boundary observable only when its minimal spinorial orbit closes inside the rank-five window. Which modes do so is not a matter of choice: the admissible catalogue is fixed by the Majorana–Fibonacci–Heegner class [20]. For each  $n \in \{2, 3, 7, 11\}$  the field  $\mathbb{Q}(\sqrt{-n})$  has class number one in the relevant phasorial channel, and its norm form defines one stable rank-one sector;  $n = 5$  is the first channel where class-one rigidity fails and the projection pole is reached. The catalogue is therefore

$$\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}, \quad (14)$$

four Heegner-stable channels and one lock-in channel [20, 21].



**Definition 3.1** (Admissibility and closure). A mode  $n \in \mathcal{N}_{\text{adm}}$  is admissible if its prime-counting index satisfies  $\pi(n) \leq r$ , where  $\pi(n)$  is the number of primes not exceeding  $n$  (so its column  $\mathbf{e}_n$  contributes to the Vandermonde rank). It is a closure node if additionally its minimal orbit length satisfies  $j_{\min}(n) \leq r$ . Here the rank- $r$  holonomy acts on  $\mathbb{Z}_n$  by the shift

$$h_r: \mathbb{Z}_n \rightarrow \mathbb{Z}_n, \quad h_r(k) = k + r \pmod{n}, \quad (15)$$

and  $j_{\min}(n)$  is the order of the orbit of 0 under  $h_r$ , i.e. the least  $j \geq 1$  with  $jr \equiv 0 \pmod{n}$ :

$$j_{\min}(n) = \frac{n}{\gcd(r, n)}. \quad (16)$$

A non-closure mode ( $j_{\min}(n) > r$ ) is confined if it lies strictly inside the window,  $\pi(n) < r$ , and saturating if it sits at the window edge,  $\pi(n) = r$ . The saturating mode is the last admissible mode, not a second confined one.

Applying (16) with  $r = 5$  [21]:

$$\begin{aligned} n = 2: & \quad \pi(2) = 1 \leq 5, \quad j_{\min} = 2/\gcd(5, 2) = 2 \leq 5 && \text{closure node,} \\ n = 3: & \quad \pi(3) = 2 \leq 5, \quad j_{\min} = 3/\gcd(5, 3) = 3 \leq 5 && \text{closure node,} \\ n = 5: & \quad \pi(5) = 3 \leq 5, \quad j_{\min} = 5/\gcd(5, 5) = 1 \leq 5 && \text{closure node (lock-in),} \\ n = 7: & \quad \pi(7) = 4 \leq 5, \quad j_{\min} = 7/\gcd(5, 7) = 7 > 5 && \text{confined,} \\ n = 11: & \quad \pi(11) = 5 \leq 5, \quad j_{\min} = 11/\gcd(5, 11) = 11 > 5 && \text{saturation.} \end{aligned} \quad (17)$$

### 3.2 Roles of the five modes

The admissibility geometry assigns each of the five modes a single definite role, summarised at the end of this section.

**$n_{\text{cov}} = 2$ , the minimal cover.** With  $j_{\min}(2) = 2 \leq r$  the mode closes inside the window. It is the smallest winding compatible with the  $4\pi$  periodicity of  $\Xi = C\bar{\Xi}$ : a  $2\pi$  rotation takes  $n = 2$  once around the internal shell  $S_\chi^2 = SU(2)/U(1)$  and closes the internal cover  $SU(2) \simeq \text{Spin}(3) \rightarrow SO(3)$  at index two. This is the double cover of the *internal* rotation group, not the Lorentz group; no spacetime  $\text{Spin}(3, 1)$  structure is invoked here. This single closed orbit carries the phase transport of the fibre and seeds the  $U(1)_\eta$  channel.

**$n_{\text{adj}} = 3$ , the adjoint seed.** Again  $j_{\min}(3) = 3 \leq r$ , so  $n = 3$  closes. It carries the adjoint label after the independently derived three-dimensional compact algebra  $\mathfrak{su}(2)_\xi$  of Section 2.3 (whose dimension is  $3 = \dim \mathfrak{su}(2)$ , one generator per independent rotation of  $S_\chi^2$ ): the winding  $n = 3$  is *assigned* to the adjoint sector, it does not by itself generate the Lie algebra. Numerically  $n_{\text{adj}} = 3$  is consistent with the three generators; the algebra itself comes from the symplectic reduction, not from the winding count.

**$n_* = 5$ , the lock-in.** Here  $j_{\min}(5) = 1$ : the mode closes at once and is a fixed point of the rank-five holonomy. It is the only mode at which the Fibonacci index reaches its own fixed point and the rank,  $F_{n_*} = n_* = r = 5$ . This fixed point is what locks the SVD filter to the boundary: the Pincherle dilatator  $D_\tau f(x) = f(\varphi x)$  acts on the logarithmic coordinate  $u = \log x$  as the unit translation  $f(u) \mapsto f(u + \log \varphi)$ . The standard degree- $\leq 4$  polynomial Mellin subspace  $V_5 = \text{span}\{(\log x)^k\}_{k=0}^4$  is invariant under this translation; the action is unipotent on  $V_5$ , with all eigenvalues equal to 1. The Fibonacci geometry of the boundary metric originates here.

**$n_{\text{conf}} = 7$ , the confined seed.** This mode is admissible and lies strictly inside the window,  $\pi(7) = 4 < r$ , but does not close:  $j_{\min}(7) = 7 > r$ . By Definition 3.1 it is confined rather than saturating. Its minimal orbit overshoots the rank, so it cannot complete an independent boundary cycle. It lives in  $\ker \Pi$ , contributes no boundary lattice element on its own, and has

zero metric weight,  $\langle \psi_7, K_\Pi \psi_7 \rangle = 0$  with  $K_\Pi = P_5^\dagger P_5$  [21]. The only way it reaches a boundary bilinear is through the cover,

$$n_{\text{conf}} \longmapsto \ell_- = n_{\text{cov}} \cdot n_{\text{conf}} = 14, \quad (18)$$

and this is forced rather than chosen: the  $2\pi$  sign of the Majorana carrier forbids the bare winding  $n = 7$  at the boundary, so it can enter only as the covered  $\ell_- = 14$  that completes the  $4\pi$  period. This is the confined kernel seed — a winding that reaches the boundary only in covered form. We call it a confined kernel seed, not a colour seed: deriving an  $SU(3)_c$  colour sector would require a further eight-generator algebra and colour dynamics, which lie outside the present scope, as does the internal structure of the confined sector in  $\ker \Pi$ .

$n_{\text{sat}} = 11$ , **saturation**. Here  $j_{\min}(11) = 11 > r$  as well, but now  $\pi(11) = 5 = r$ : the mode sits at the window edge, where any larger prime would need  $\pi(p) \geq 6$  and fall outside. By Definition 3.1 the edge value  $\pi(n) = r$  makes  $n = 11$  the saturating mode — the last admissible mode and the upper lever of the slab — not a second confined one.

The three closure nodes  $D = \{n_{\text{cov}}, n_{\text{adj}}, n_*\} = \{2, 3, 5\}$  are the boundary's local spinor–adjoint vocabulary: their orbits close inside the window, and they alone set the gauge content the boundary can absorb.

| Mode                  | Physical role                                        | Boundary reading                 | Status       |
|-----------------------|------------------------------------------------------|----------------------------------|--------------|
| $n_{\text{cov}} = 2$  | Minimal spinorial cover, $4\pi$ closure              | Abelian $U(1)_\eta$ channel      | closure node |
| $n_{\text{adj}} = 3$  | First non-Abelian adjoint seed                       | $SU(2)_\xi$ seed                 | closure node |
| $n_* = 5$             | Fibonacci lock-in, $F_{n_*} = n_* = r$               | SVD lock-in sector               | closure node |
| $n_{\text{conf}} = 7$ | $j_{\min}(7) > r$ , $\pi(7) < r$ : inside the window | confined kernel seed, $\ker \Pi$ | confined     |
| $n_{\text{sat}} = 11$ | $j_{\min}(11) > r$ , $\pi(11) = r$ : window edge     | upper lever                      | saturating   |

## 4 Forcing of the saturation mode and the generator norm

This section shows that the spinorial generator and its norm are fixed, under the adopted covering convention, by the structure of the rank-five admissible catalogue: a saturation argument on the prime-counting function determines which modes are distinguished, and the generator is built from them. The two integers entering the generator follow from this saturation under the adopted covering convention; their norm is the resulting structural invariant. The further extraction of the residue from that norm is treated separately and is *not* closed by the present argument; the scope here is limited to the generator and its norm.

### 4.1 Admissible prime denominators

The admissible angular catalogue of Section 3 is restricted to prime denominators,

$$\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}, \quad (19)$$

the primes up to the saturation point fixed below. (The restriction to primes is the catalogue convention of Section 3; we use it here without re-deriving it.) For a denominator  $n$ , the relevant count is the number of admissible (prime) resolutions not exceeding  $n$ , i.e. the prime-counting function  $\pi(n)$ .



## 4.2 The prime lattice and its renormalised fixed point

That  $\pi(n)$  is the correct counting function — rather than  $n$  itself — is an intrinsic property of the prime lattice, made precise by the following limit. Let  $p_k$  be the  $k$ -th prime and define on  $S^1(\tau)$  the prime-only angular sequence  $\theta_k = \pi/p_k$  with its density-renormalised version

$$\Phi_k := (k \ln k) \frac{\pi}{p_k}. \quad (20)$$

**Theorem 4.1** (Prime-lattice fixed point).  $\lim_{k \rightarrow \infty} \Phi_k = \pi$ .

*Proof.* By Cipolla's asymptotic expansion of the  $k$ -th prime [14],

$$p_k = k \ln k + k \ln \ln k - k + O\left(\frac{k \ln \ln k}{\ln k}\right), \quad (21)$$

so substituting in (20),

$$\Phi_k = \frac{\pi}{1 + \frac{\ln \ln k}{\ln k} - \frac{1}{\ln k} + O\left(\frac{\ln \ln k}{\ln^2 k}\right)} \xrightarrow{k \rightarrow \infty} \pi, \quad (22)$$

since  $\ln \ln k / \ln k \rightarrow 0$  and  $1 / \ln k \rightarrow 0$ . (Convergence from below,  $\Phi_k < \pi$ , would follow from the lower bound  $p_k > k \ln k$  of Rosser's theorem; we do not use the inequality here, only the limit.)  $\square$

The renormaliser  $k \ln k$  in (20) is the leading prime-counting estimate  $p_k \sim k \ln k$  (equivalently  $\pi(n) \sim n / \ln n$ ): the density of admissible resolutions at scale  $n$  is  $1 / \ln n$ , and its cumulative count is  $\int^n dt / \ln t \sim n / \ln n \sim \pi(n)$ . Theorem 4.1 is consistent with the prime-counting parametrisation used in the admissibility catalogue of Section 3; it does not independently derive that catalogue, only exhibit the fixed point  $\pi$  of the density renormalisation.

## 4.3 Saturation at $n = 11$

The rank bound is  $\pi(n) \leq r = 5$ , with  $r$  fixed by the singular-value threshold  $w_4 > \alpha > w_5$  of Section 3, independently of the present argument. The bound saturates at  $n = 11$ :

$$\pi(11) = 5 = r, \quad \pi(13) = 6 > r, \quad (23)$$

so  $n_{\text{sat}} = 11$  is the largest admissible denominator and  $n = 13$  the first excluded one. The cutoff is robust to the analytic form of the count: using the estimate  $n / \ln n$  in place of the exact  $\pi(n)$ ,

$$\frac{11}{\ln 11} = 4.59 < 5 < 5.07 = \frac{13}{\ln 13}, \quad (24)$$

so the bound  $n / \ln n \leq 5$  on primes selects the same pair — 11 admitted, 13 excluded — as the exact statement (23). The saturation node is forced by the integer-exact relation (23); the analytic estimate is a consistency check, not the source of the value.

## 4.4 The generator and its norm

Two admissible modes are distinguished by failing the closure test  $j_{\min}(n) = n / \gcd(5, n) \leq r$  of Section 3: the confined mode  $n = 7$  ( $j_{\min} = 7 > 5$ ) and the saturation mode  $n = 11$  ( $j_{\min} = 11 > 5$ ). The spinorial generator pairs the saturation mode with twice the confined mode,

$$v_{\text{spin}} = (n_{\text{sat}}, 2n_{\text{conf}}) = (11, 14), \quad \gcd(11, 14) = 1, \quad (25)$$

the factor 2 being the spinorial doubling of the carrier. (The covering rule  $(n_{\text{sat}}, 2n_{\text{conf}})$  is the one used here; it is not shown to be the unique admissible pairing of the two distinguished modes — the alternative  $(2n_{\text{sat}}, n_{\text{conf}})$  is not excluded.) Its norm is the orthogonal-closure invariant of the cell spanned by  $v_{\text{spin}}$  and  $v_{\text{spin}}^\perp = (-14, 11)$ ,

$$N_{\text{spin}} = \det \begin{pmatrix} 11 & -14 \\ 14 & 11 \end{pmatrix} = 11^2 + 14^2 = 317. \quad (26)$$

Both integers in (25), and hence the norm (26), are fixed by the admissible catalogue.

**Remark 4.1** (Scope of the closure). *Under the adopted covering convention the generator  $v_{\text{spin}} = (11, 14)$  is fixed, with norm  $N_{\text{spin}} = 317$ , from the saturation (23) and the confined mode. This norm enters neither principal result: the internal ratio  $\Theta_\partial$  of Section 6.2 uses only  $r$ , the chiral doubling, and the channel count  $|\mathcal{C}_{\text{obs}}|$  of Section 9, and the fine-structure scale of Section 5 uses only the Pincherle increment and the Fibonacci curvature. The norm 317 serves only as an arithmetic label of the invariant Abelian residue in Section 11.*

## 5 A Pincherle construction of the projection-pole scale

The projection threshold  $\alpha$  is the inverse square of a projection scale  $\Lambda$  built structurally from the Pincherle dilation operator  $\mathcal{D} = x D_x$  acting on the  $\varphi$ -autosimilar hierarchy, and  $\alpha^{-1} = \Lambda^2$  is the pole of the boundary transfer filter [21]. The construction of  $\Lambda$  is given first; its numerical value, and the integer floor it yields, are recorded only afterwards, as a readout of that structure rather than an arithmetic residue.

### 5.1 The Pincherle projection scale

Let  $\mathcal{D} = x d/dx$  be the Pincherle dilation generator acting on the  $\varphi$ -autosimilar scale hierarchy. Two structural invariants are available, each fixed independently of any rank or threshold:

- (i) the elementary bulk scale increment at the quadratic autosimilar level  $x^2$ ,

$$\mathcal{D}(x^2)|_{x=\varphi} = 2\varphi^2; \quad (27)$$

- (ii) the spectral curvature of the Fibonacci recursion,  $\kappa = \sqrt{5}$ , fixed by the discriminant  $\Delta = (\lambda_+ - \lambda_-)^2 = 5$  of  $\varphi^2 = \varphi + 1$  [21].

The bulk-to-boundary projection scale is their product,

$$\Lambda := \mathcal{D}(x^2)|_{x=\varphi} \cdot \kappa = 2\varphi^2\sqrt{5}, \quad (28)$$

introducing no independent numerical coefficient. Using  $\sqrt{5} = 2(\varphi - \frac{1}{2})$  and  $\varphi^2 = \varphi + 1$ , the scale expands in  $\mathbb{Q}(\sqrt{5})$  as

$$\Lambda = 4\varphi^2(\varphi - \frac{1}{2}) = F_5 + F_4\sqrt{5} = 5 + 3\sqrt{5}, \quad (29)$$

the  $\mathbb{Q}(\sqrt{5})$ -coordinates of (28) being the consecutive Fibonacci numbers  $F_5, F_4$ , not an independent selection, with 5, 3 their numerical readout. The coordinates  $F_5, F_4$  here are the *output* of the golden identity applied to the Pincherle increment, not posited. Equation (28) is the projection scale built from the two structural invariants (27) and  $\kappa$ ; we do not claim a uniqueness theorem beyond this construction.

## 5.2 The threshold and the pole

The projection threshold is the inverse square of the scale,

$$\alpha := \Lambda^{-2} = [4\varphi^2(\varphi - \tfrac{1}{2})]^{-2} = \frac{1}{20\varphi^4} = \frac{7 - 3\sqrt{5}}{40} \approx 7.295 \times 10^{-3}, \quad (30)$$

a number in  $\mathbb{Q}(\sqrt{5})$  depending only on  $\varphi$ . The boundary transfer filter of the rank-five projection is the one-pole rational function

$$H(z) = \frac{P_4(z)}{1 - \alpha z}, \quad \deg P_4 = 4, \quad (31)$$

where  $\deg P_4 = 4$  is forced by rank five and the single pole by the single projection scale [21]. Its pole sits at  $z_{\text{pole}} = \alpha^{-1}$ , the reciprocal threshold,

$$\boxed{\alpha^{-1} = \Lambda^2 = (F_5 + F_4\sqrt{5})^2 = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5} \approx 137.082.} \quad (32)$$

The floor of this irrational pole,  $\lfloor 70 + 30\sqrt{5} \rfloor = 137$ , is the spectral residue associated with the distinguished Abelian Cartan channel of Section 6 (its electromagnetic identification belongs to M3). Unlike the arithmetic residue,  $\alpha^{-1}$  here is not an integer: it is the irrational pole  $137.08\dots$ , of which 137 is the integer part.

**Remark 5.1** (What the pole route establishes). *The pole route (32) returns the irrational  $\alpha^{-1} = 137.08\dots$  and depends only on the Pincherle increment  $2\varphi^2$  and the Fibonacci curvature  $\sqrt{5}$ . It is the quantity that admits a physical comparison:  $\alpha^{-1} = 137.082$  versus the measured 137.036 [21]. What is not established here is a uniqueness theorem for  $\Lambda$  (only its construction from the two structural invariants), nor the precise origin of the small offset  $137.082 - 137.036$  from the measured value, which is a physical question.*

## 5.3 Spinorial reading of the increment factor (conditional)

The scale (28) carries two numerical factors: the 2 of the Pincherle increment (27) and the  $\sqrt{5}$  of the curvature. Both admit a reading in which they originate from the Majorana structure of the carrier rather than from the scale hierarchy alone. The reading is conditional, and its hypotheses are stated explicitly.

**Proposition 5.1** (Chiral origin of the increment factor). *Assume*

- (P1) *the boundary observables are the Majorana bilinears  $\bar{\Xi}\Gamma\Xi$  of (3);*
- (P2) *the fundamental Majorana mode sits at the linear autosimilar scale level, i.e. it carries Pincherle weight  $s_0 = 1$  under  $\mathcal{D} = x d/dx$  (equivalently  $s_0 = 2n$  with the lowest antiperiodic Fourier index  $n = \frac{1}{2}$ ).*

*Then the factor 2 in  $\Lambda = 2\varphi^2\sqrt{5}$  of (28) is the degree of the Majorana bilinear, i.e. the chiral doubling carried by the double sheet  $\Xi(\chi + 2\pi) = -\Xi(\chi)$ .*

*Proof.* By the double-sheet condition  $\Xi(\chi + 2\pi) = -\Xi(\chi)$ , a boundary quantity is single-valued under  $\chi \rightarrow \chi + 2\pi$  only if it is even in  $\Xi$ ; the lowest such quantities are the bilinears  $\bar{\Xi}\Gamma\Xi$ , since  $(-\Xi)(-\Xi) = +\Xi\Xi$ . By (P1) these are the observables. Under  $\mathcal{D}$  a bilinear carries twice the scale weight of the spinor: if  $\Xi \sim x^{s_0}$  then  $\bar{\Xi}\Gamma\Xi \sim x^{2s_0}$ , the factor 2 being the bilinear degree. By (P2)  $s_0 = 1$ , so the observable sits at the quadratic level  $x^2$ , with elementary increment  $\mathcal{D}(x^2)|_{x=\varphi} = 2\varphi^2$  as in (27). The factor 2 in (28) is therefore the bilinear degree, carried by the double sheet.  $\square$

**Remark 5.2** (The curvature is the Majorana-pair discriminant). *The second factor admits an analogous reading. The Majorana transport pair acquires the spinorial sign on the inverse traversal,  $(\lambda, -\lambda^{-1})$ ; at  $\lambda = \varphi$  it satisfies*

$$\mu^2 - (\varphi - \varphi^{-1})\mu - 1 = \mu^2 - \mu - 1 = 0,$$

*the golden equation, with discriminant  $\Delta = 5$  and root separation  $\sqrt{\Delta} = \sqrt{5} = \kappa$ . The bosonic pair  $(\varphi, +\varphi^{-1})$  has  $\Delta = 1$  instead. Thus, under Proposition 5.1, both numerical factors of  $\Lambda = 2\varphi^2\sqrt{5}$  trace to the spinorial (minus-sign) structure: the 2 as the bilinear degree, the  $\sqrt{5}$  as the discriminant separation of the Majorana pair.*

**Remark 5.3** (Status of (P2)). *Hypothesis (P2) cannot be derived from the antiperiodicity alone. The antiperiodic condition constrains the chiral phase index  $n = \frac{1}{2}$ , which lives on the fibre  $\chi$ ; the Pincherle weight  $s_0$  lives on the scale variable  $x$  and is the eigenvalue of a different operator. The relation  $s_0 = 2n$  that links them is itself the chiral doubling: imposing the alternative identification  $s_0 = n$  would place the observable at the linear level  $x^1$ , not the quadratic level. The factor 2 is therefore a primitive structural input, on the same footing as the spinorial sign, and not a consequence of the antiperiodicity. Proposition 5.1 changes only the reading of this factor; the value  $\Lambda^2 = 20\varphi^4 \approx 137.08$  of (32) is unaffected.*

## 6 The internal dimension ratio

The rank, the chiral doubling, and the observable channel count now combine into the internal dimension ratio. The construction uses no block constant and no arithmetic split of the generator norm: the generator  $v_{\text{spin}} = (11, 14)$  and its norm  $N_{\text{spin}} = 317$  are fixed in Section 4, and the fine-structure scale  $\alpha^{-1}$  in Section 5; neither enters the ratio below.

### 6.1 The three observable forms

The boundary doublet carries  $\mathfrak{su}(2)_\xi$  (Section 7), fixed by the metric degeneracy of the boundary ( $USp(2) \cong SU(2)$ ) and independently reproduced by the Fibonacci braid envelope (Theorem 9.4). Its adjoint sector has dimension

$$|\mathcal{C}_{\text{obs}}| = \dim \mathfrak{su}(2) = 3, \quad (33)$$

giving three observable forms: the raising/lowering pair  $T_\pm$  and the Cartan generator  $T_3$ . The identification of  $(T_\pm, T_3)$  with the electroweak triplet  $(W^-, W^+, Z^0)$  is the conditional bridge of Section 7, not assumed here.

### 6.2 The internal dimension ratio

**Definition 6.1** (Internal dimension ratio). *The internal dimension ratio of the rank-five Majorana boundary is the fraction of the internal dimension  $2r + |\mathcal{C}_{\text{obs}}|$  occupied by the observable adjoint channels,*

$$\Theta_\partial := \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|}, \quad (34)$$

*where  $2r$  is the chiral double sheet over the  $r$  retained modes (Proposition 6.3) and  $|\mathcal{C}_{\text{obs}}|$  is the adjoint channel count.*

This is a ratio of dimensions, not a quantity derived from a Hessian, a measure, or a mixing dynamics; the name “mixing ratio” is retained only as the internal interpretation, whose identification with a kinetic mixing angle requires the conditions M1–M3 of Section 7.

**Corollary 6.2** (Value of the ratio). *Each ingredient of (34) is fixed structurally:  $r = 5$  by the singular-value threshold  $w_4 > \alpha > w_5$  (Section 3); the factor 2 by the Majorana double cover; and  $|\mathcal{C}_{\text{obs}}| = 3$  by the Majorana symplectic doublet of Section 2.3 ( $USp(2) \cong SU(2)$ ), with Theorem 9.4 reproducing the same count under (H1)–(H3). The value  $3/13$  thus does not depend on the still-open categorical selection. Hence*

$$\boxed{\Theta_{\partial} = \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|} = \frac{3}{2 \cdot 5 + 3} = \frac{3}{13}.} \quad (35)$$

The value  $3/13$  uses only the rank, the chiral doubling, and the channel count; it introduces no free parameter, coupling constant, or symmetry-breaking mechanism. Its identification with the tree-level electroweak mixing angle,  $\Theta_{\partial} = \sin^2 \theta_W$ , is the conditional bridge of Section 7: it requires deriving  $g'^2/g^2$  from the normalisation of the projected currents, which is not done here.

### 6.3 The internal dimension 13

The denominator 13 in (35) is the dimension of a boundary subspace  $\mathcal{B}_{13} = S_5 \oplus T_8$ . This space carries two structural decompositions — the ribbon decomposition  $S_5 \oplus T_8$  and the Hardy block decomposition into the  $P_+$  and  $P_-$  sectors of the chiral fibre  $S_{\chi}$  — whose numerical readouts are  $5 + 8$  and  $10 + 3$ .

**Proposition 6.3** (Hardy origin of  $\dim \mathcal{B}_{13} = 13$ ). *Let  $\mathcal{H}_{\text{Cay}}$  be the Hilbert–Cayley operator on the chiral fibre  $S_{\chi}$ , satisfying*

$$\mathcal{H}_{\text{Cay}}^2 = -I, \quad \mathcal{H}_{\text{Cay}}^{\dagger} = -\mathcal{H}_{\text{Cay}}, \quad \text{spec}(i\mathcal{H}_{\text{Cay}}) = \{+1, -1\},$$

*so that  $i\mathcal{H}_{\text{Cay}}$  is self-adjoint with eigenvalues  $\pm 1$ , each of multiplicity one. Let*

$$P_{\pm} = \frac{1}{2}(I \pm i\mathcal{H}_{\text{Cay}})$$

*be the corresponding rank-one Hardy projectors, so  $P_+P_- = 0$ ,  $P_+ + P_- = I$ ,  $\text{rank } P_{\pm} = 1$ , and  $S_{\chi} = P_+S_{\chi} \oplus P_-S_{\chi}$  with  $\dim S_{\chi} = 2$ . Let  $A_4$  be the abstract four-dimensional gauge module of the boundary,*

$$A_4 = \mathbb{C}a_0 \oplus A_3, \quad A_3 = \text{span}\{a_1, a_2, a_3\},$$

*where  $a_0$  is a scalar singlet readout associated with the Cartan residue — the trace (identity) direction of the readout module, not an independent gauge generator, since the Abelian Cartan  $U(1)_{\eta}$  is already one of  $a_1, a_2, a_3$  — and  $a_1, a_2, a_3$  are the three adjoint generators of Section 7 (identification with physical gauge fields is the conditional bridge of that section and is not assumed here). Then the tensor sector carries the polarisation doublet of  $S_{\chi}$ :*

$$T_8 = A_4 \otimes S_{\chi} = (\mathbb{C}a_0 \otimes S_{\chi}) \oplus (A_3 \otimes P_+S_{\chi}) \oplus (A_3 \otimes P_-S_{\chi}), \quad \dim T_8 = 2 + 3 + 3 = 8, \quad (36)$$

*and the boundary ribbon space decomposes as*

$$\mathcal{B}_{13} = S_5 \oplus T_8 = [S_5 \oplus (\mathbb{C}a_0 \otimes S_{\chi}) \oplus (A_3 \otimes P_+S_{\chi})] \oplus (A_3 \otimes P_-S_{\chi}), \quad (37)$$

*with*

$$\dim \mathcal{B}_{13} = \dim S_5 + \dim T_8 = 5 + 8 = 13, \quad (38)$$

*the dimension of the direct sum  $\mathcal{B}_{13} = S_5 \oplus T_8$  being the sum of the dimensions of its two summands.*

*Proof. Hardy projectors.* Since  $\mathcal{H}_{\text{Cay}}$  is anti-self-adjoint with  $\mathcal{H}_{\text{Cay}}^2 = -I$ , the operator  $A := i\mathcal{H}_{\text{Cay}}$  is self-adjoint with  $A^2 = I$ , hence  $\text{spec}(A) = \{+1, -1\}$ . With  $\dim S_\chi = 2$  each eigenvalue has multiplicity one, so  $\text{rank } P_\pm = 1$ , and  $P_\pm = \frac{1}{2}(I \pm A)$  satisfy  $P_\pm^2 = P_\pm$ ,  $P_+P_- = 0$ ,  $P_+ + P_- = I$ .

*Dimension of  $T_8$ .* Four abstract channels  $(a_0, a_1, a_2, a_3)$ , each tensored with  $S_\chi$  ( $\dim S_\chi = 2$ ): total  $4 \times 2 = 8$ . The split into  $2 = \dim(\mathbb{C}a_0 \otimes S_\chi)$  and  $3+3 = \dim(A_3 \otimes P_\pm S_\chi)$  gives  $2+3 \times 2 = 8$ . The factor 2 multiplying  $A_3$  is the chiral doubling, and  $|A_3| = 3 = |\mathcal{C}_{\text{obs}}|$  is the channel count of Theorem 9.4.

*Decomposition of  $\mathcal{B}_{13}$ .*  $S_5 = \text{span}\{(\log x)^k\}_{k=0}^4$  has dimension 5 [21]. The block  $[S_5 \oplus (\mathbb{C}a_0 \otimes S_\chi) \oplus (A_3 \otimes P_+ S_\chi)]$  has dimension  $5 + 2 + 3 = 10$ ; the complementary block  $(A_3 \otimes P_- S_\chi)$  has dimension 3, so the two Hardy blocks have dimensions 10 and 3, summing to  $\dim \mathcal{B}_{13} = 13$ . This  $10 + 3$  Hardy split is recorded only as a regrouping: the ten-dimensional block above is not identified here with a canonical double sheet  $V_5 \otimes S_\chi$  (that would require an explicit isomorphism, not constructed). The structural  $2r + |\mathcal{C}_{\text{obs}}|$  form of the denominator belongs to the counting space  $\mathcal{B}_{\text{count}}$  (Definition 6.4), not to  $\mathcal{B}_{13}$ .  $\square$

**Remark 6.1.** *The two factors of 2 used here — the longitudinal/transverse polarisation doublet of each gauge channel and the Hardy split  $S_\chi = P_+ S_\chi \oplus P_- S_\chi$  — are two readings of the same chiral fibre, generated by the single operator  $\mathcal{H}_{\text{Cay}}$ . The denominator of  $\Theta_\partial$  is  $\dim \mathcal{B}_{\text{count}} = 13$  (Definition 6.4); the ribbon  $\mathcal{B}_{13}$  realises the same dimension. The numerator is the channel count  $|\mathcal{C}_{\text{obs}}| = 3$ , giving  $\Theta_\partial = 3/13$ .*

The canonical  $2r + |\mathcal{C}_{\text{obs}}|$  form of the denominator is carried by a graded space built directly from the rank-five chiral double sheet and the adjoint algebra, independently of the ribbon  $\mathcal{B}_{13}$ .

**Definition 6.4** (Counting space). *Let  $V_5$  be the rank-five modal space ( $\dim V_5 = r = 5$ , Section 3),  $S_\chi$  the chiral doublet ( $\dim S_\chi = 2$ ), and  $\mathfrak{su}(2)_\xi$  the boundary adjoint algebra of Section 2.3. The counting space is*

$$\mathcal{B}_{\text{count}} := (V_5 \otimes S_\chi) \oplus \mathfrak{su}(2)_\xi, \quad \dim \mathcal{B}_{\text{count}} = 2r + 3 = 13, \quad (39)$$

*the chiral double sheet  $V_5 \otimes S_\chi$  of dimension  $2r = 10$  plus the three adjoint generators.*

In  $\mathcal{B}_{\text{count}}$  the split  $13 = 2r + |\mathcal{C}_{\text{obs}}|$  is canonical by construction: the  $2r$  is literally the chiral double sheet over the five modes and the 3 is literally  $\dim \mathfrak{su}(2)_\xi$ , so Definition 6.1 applies to it directly. The ribbon  $\mathcal{B}_{13} = S_5 \oplus T_8$  of Proposition 6.3 is a dimensionally compatible realisation,  $\dim \mathcal{B}_{13} = \dim \mathcal{B}_{\text{count}} = 13$ ; we do not identify the two spaces, as that would require an explicit isomorphism not constructed here. The ratio  $\Theta_\partial$  is defined on  $\mathcal{B}_{\text{count}}$ , where the decomposition is canonical.

## 7 The adjoint structure and the electroweak bridge

The internal ratio  $\Theta_\partial = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|) = 3/13$  is fixed. Connecting it to the physical electroweak sector involves two steps of very different status: the adjoint Lie structure, which closes here, and a set of kinetic and factor-identification conditions, which do not.

**The adjoint Lie algebra from the metric degeneracy.** The  $\mathfrak{su}(2)$  structure has a single origin, read in four logical steps, each acting on a different object. It begins with the projection metric, not with the fusion category.

(1) *The projection selects the doublet.* The chiral doublet is the degenerate singular subspace of the boundary metric  $K_\Pi = \Pi^\dagger \Pi$ ,

$$S_\chi \equiv E_\chi := \ker(K_\Pi - \sigma_\chi^2 I), \quad \dim_{\mathbb{C}} E_\chi = 2, \quad K_\Pi|_{E_\chi} = \sigma_\chi^2 I_2. \quad (40)$$



Because the metric is scalar on  $E_\chi$ , its full isometry group is  $G_K = U(E_\chi) \cong U(2)$ . This is the operator-theoretic content of a finite-rank boundary metric: a degenerate singular doublet carries a  $U(2)$  stabiliser, conditional on the spectral partition that produces the degeneracy.

(2) *The Majorana reality restricts  $U(2)$  to  $SU(2)$ .* Removing the central phase does *not* give  $SU(2)$  globally: as a group  $U(2) \cong (U(1) \times SU(2))/\mathbb{Z}_2$ , with  $\mathfrak{u}(2) = \mathfrak{u}(1) \oplus \mathfrak{su}(2)$ . The reduction is supplied by the Majorana structure: on the doublet the self-conjugacy  $\Xi = C\bar{\Xi}$  preserves a skew form  $\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ , and the admissible group is the compact symplectic group

$$G_\chi = \{U \in U(2) : U^T \varepsilon U = \varepsilon\} = USp(2) \cong SU(2), \quad (41)$$

with Lie algebra  $\mathfrak{g}_\chi = \{X : X^\dagger K_\Pi + K_\Pi X = 0, X^T \varepsilon + \varepsilon X = 0\} \cong \mathfrak{su}(2)$ . By Schur's lemma the commutant of this irreducible  $\mathfrak{su}(2)$  in  $\text{End}(E_\chi)$  is  $\mathbb{C}I_2$ , and the only continuous generator outside  $\mathfrak{su}(2)$ , the central phase  $iI_2$ , violates the symplectic condition,  $(iI_2)^T \varepsilon + \varepsilon(iI_2) = 2i\varepsilon \neq 0$ . No independent Abelian factor therefore survives the Majorana reality:  $\mathfrak{g}_\chi \cong \mathfrak{su}(2)$  is the *complete* stabiliser, and the real rotation  $U(1)_\eta$  of Section 2.2 sits inside it as a Cartan, not as a direct summand.

(3) *The endomorphism algebra realises  $\mathfrak{su}(2)$ .* The traceless endomorphisms of the doublet give the explicit Pauli realisation

$$\text{End}_0(E_\chi) \cong \mathfrak{sl}(2, \mathbb{C}), \quad T_a = \frac{1}{2}\sigma_a, \quad [T_a, T_b] = i\varepsilon_{abc}T_c, \quad (42)$$

whose compact real form is the  $\mathfrak{su}(2)$  of step (2) — a realisation of the same Lie algebra, not a second independent origin.

(4) *The anisotropy orients the Cartan.* Steps (1)–(3) use only the scalar metric  $K_\Pi|_{E_\chi} = \sigma_\chi^2 I_2$ . The anisotropic tensor  $A_\chi = \text{diag}(\alpha_\parallel, \alpha_\perp)$  with  $\alpha_\parallel \neq \alpha_\perp$  is a *separate* operator on  $E_\chi$  (not the metric), and selects a Cartan generator

$$T_3 \propto A_\chi - \frac{1}{2}(\text{tr } A_\chi) I_2 = \frac{1}{2}(\alpha_\parallel - \alpha_\perp) \sigma_3, \quad (43)$$

with  $T_\pm$  the eigenvectors of its adjoint action,  $[T_3, T_\pm] = \pm T_\pm$ . The degeneracy generates the orientation freedom; the anisotropy chooses the axis within it. There is no tension between  $K_\Pi|_{E_\chi} \propto I_2$  and  $A_\chi$  anisotropic: they are different operators acting at different logical stages.

In one line:

SVD selects the doublet  $\rightarrow$  Majorana selects  $SU(2) \rightarrow$   $\text{End}_0$  realises  $\mathfrak{su}(2) \rightarrow A_\chi$  orients  $T_3$ .

**Generators, Cartan alignment, and the global stabiliser.** The stabiliser  $\mathfrak{g}_\chi$  is the anti-Hermitian algebra spanned by  $t_a = -\frac{i}{2}\sigma_a$ , with real structure constants  $[t_a, t_b] = \varepsilon_{abc}t_c$ ; the Hermitian Pauli generators of step (3) are  $T_a = it_a$ . The real rotation  $U(1)_\eta$  of Section 2.2 is the one-parameter subgroup generated by  $J_\eta = -i\sigma_2 = 2t_2$ , a Cartan of  $\mathfrak{su}(2)$ . It is conjugate to the anisotropy axis  $T_3$  of step (4) by the internal rotation  $W = \exp(-\frac{i\pi}{4}\sigma_1)$ , for which  $W\sigma_2W^{-1} = \sigma_3$ ; fixing the Pauli basis aligned to the anisotropy,  $U(1)_\eta$  and the unbroken Cartan  $T_3$  are the same maximal torus.

The definition (4) fixes  $\mathfrak{g}_\partial$  as the stabiliser of the observable forms on all of  $\mathcal{H}_{\text{obs}}$ , not only on the doublet  $E_\chi$ ; the two coincide. Any  $X \in \mathfrak{g}_\partial$  is anti-Hermitian and preserves the boundary metric,  $X^\dagger K_\Pi + K_\Pi X = 0$ , i.e.  $[K_\Pi, X] = 0$ . Since  $K_\Pi$  carries the degenerate value  $\sigma_\chi^2$  on  $E_\chi$  and *distinct* singular values on the remaining one-dimensional eigenspaces, commutation forces  $X$  block-diagonal: it cannot mix eigenspaces of different eigenvalue, and on each non-degenerate line it acts as an imaginary scalar  $i\theta_k$ . The Majorana reality removes every such continuous phase — on a self-conjugate line  $e^{i\theta_k}$  forces  $e^{2i\theta_k} = 1$ , leaving only  $\mathbb{Z}_2$  — so the one-dimensional blocks contribute no continuous generator. The sole continuous freedom is the  $\mathfrak{u}(2)$  on the degenerate doublet, reduced by  $\varepsilon$  to  $\mathfrak{su}(2)$ , which gives  $\mathfrak{g}_\partial \subseteq \mathfrak{su}(2)_\varepsilon$ .

For the reverse inclusion, the observable forms restricted to the doublet are  $\mathcal{B}_\partial|_{E_\chi} = \text{span}\{K_\chi, \varepsilon\}$  — the scalar metric  $K_\chi = \sigma_\chi^2 I_2$  and the Majorana symplectic form  $\varepsilon$ . Every

$X \in \mathfrak{su}(2)_\xi$  preserves both:  $X^\dagger K_\chi + K_\chi X = \sigma_\chi^2(X^\dagger + X) = 0$  for anti-Hermitian  $X$ , and  $X^T \varepsilon + \varepsilon X = 0$  for traceless  $X$ . Hence  $\mathfrak{su}(2)_\xi \subseteq \mathfrak{g}_\partial$ , and

$$\mathfrak{g}_\partial = \mathfrak{g}_\chi \cong \mathfrak{su}(2)_\xi, \quad (44)$$

the chiral doublet  $E_\chi$  carrying the entire continuous gauge content of the boundary.

The anisotropy  $A_\chi = \text{diag}(\alpha_\parallel, \alpha_\perp)$  of step (4) is *not* an additional fixed form in this stabiliser: it enters covariantly, transforming in the adjoint  $A_\chi \mapsto U A_\chi U^{-1}$  ( $U \in SU(2)_\xi$ ) with non-trivial orbit. It is an order parameter that orients — spontaneously, as a background direction — the unbroken Cartan  $U(1)_\eta \subset \mathfrak{su}(2)_\xi$ , breaking the gauge algebra to its maximal torus without removing any generator from it. Were  $A_\chi$  imposed as a fixed observable, the stabiliser would collapse to  $\text{Stab}_{SU(2)}(A_\chi) = U(1)$ ; the metric degeneracy  $K_\Pi|_{E_\chi} \propto I_2$  is precisely the statement that the gauge-defining form carries no anisotropy, so  $\mathfrak{g}_\partial$  is the full  $\mathfrak{su}(2)_\xi$  and the anisotropy acts on it, not within its defining constraint.

The abstract Temperley–Lieb category of Section 8 does not by itself determine the boundary gauge Lie algebra. However, its irreducible two-dimensional unitary Fibonacci braid representation has full matrix envelope  $M_2(\mathbb{C})$  (Proposition 9.3), whose traceless commutator algebra is  $\mathfrak{sl}_2(\mathbb{C})$  and whose compact real form is  $\mathfrak{su}(2)$  (Theorem 9.4). The Majorana structure identifies the compact form, while the carrier anisotropy selects its Cartan axis. The earlier obstruction — that the associative fusion representation  $\text{TL}_4(\varphi)$  cannot map canonically to a Lie algebra — is thereby resolved: the braid envelope realises the same compact Lie algebra obtained from the symplectic doublet in Section 2.3, and the channel count  $|\mathcal{C}_{\text{obs}}| = 3$  is fixed there and reproduced in Section 9 under (H1)–(H3), subject to the intertwiner of Remark 9.1.

**Conditional electroweak kinetic matching.** The odds ratio associated with the internal dimension ratio is

$$\frac{\Theta_\partial}{1 - \Theta_\partial} = \frac{3/13}{10/13} = \frac{3}{10}. \quad (45)$$

This odds ratio would become the physical coupling ratio  $g'^2/g^2$  only under a *conditional collective-adjoint ansatz*: that the gauge Hessian induced by the projection takes the form

$$\Gamma_{\text{gauge}}^{(2)} = -\frac{c}{4} \left( \mathcal{I}_1 F_{\mu\nu}^Y F_Y^{\mu\nu} + \mathcal{I}_2 F_{\mu\nu}^a F_a^{\mu\nu} \right), \quad \mathcal{I}_1 = \dim V_5 = 5, \quad \mathcal{I}_2 = \text{tr}_{S_\chi} \sum_a T_a^2 = \frac{3}{2}. \quad (46)$$

This is not the standard Yang–Mills kinetic normalisation. In the standard per-generator convention  $\text{tr}(T_a T_b) = \frac{1}{2} \delta_{ab}$  one has, per generator,

$$H_{ab} = \frac{c}{2} \delta_{ab}, \quad \text{whereas} \quad \sum_{a=1}^3 H_{aa} = \frac{3c}{2} \quad (47)$$

is the collective trace. The matching to  $\sin^2 \theta_W = 3/13$  therefore requires three separate conditions, none established here:

**M1** (block uniformity) the projection metric is block-uniform on the rank-five sector,

$$K_\Pi|_{\mathcal{K}_{10}} = c K_{\text{FMT}}^{(5)} \otimes I_\chi,$$

where  $K_{\text{FMT}}^{(5)} \in \text{End}(V_5)$  is the rank-five Fibonacci–Mellin frame metric on the five-dimensional logarithmic block [23], and  $I_\chi$  acts on the two-dimensional chiral fibre, so the product acts on  $\mathcal{K}_{10} = V_5 \otimes S_\chi$ ;

**M2** (collective-adjoint matching) the physical  $SU(2)$  kinetic coefficient equals the collective trace,  $Z_W^{\text{phys}} = \sum_{a=1}^3 H_{aa} = \frac{3c}{2}$ ;

**M3** (factor identification) since the real rotation  $U(1)_\eta$  is the Cartan of  $SU(2)_\xi$ , not an independent factor, the hypercharge  $U(1)_Y$  is *not*  $U(1)_\eta$ : it requires an independent Abelian sector on an additional space, absent from the present Majorana doublet. The remaining conditions are the identification  $SU(2)_\xi = SU(2)_L$ , the left-chiral action on the doublet, the hypercharge normalisation  $Y^\dagger Y = I_{V_5}$  on that new sector, and the absence of kinetic mixing between it and  $SU(2)_\xi$ .

M1 alone does *not* imply  $Z_\xi = \frac{3c}{2}$ : that value follows only from M1 together with M2. A Schur argument gives one constant on each irreducible block but permits two distinct constants; M1 fixes them equal, and M2 then matches the collective trace to the physical coefficient.

**Status.**

$$\Theta_\partial = \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|} = \frac{3}{13} \quad (\text{internal theorem}), \quad (48)$$

while  $\Theta_\partial = \sin^2 \theta_W$  remains conditional on M1–M3. *The internal ratio 3/13 is closed within the boundary algebra. Its identification with the electroweak coupling ratio remains conditional on M1–M3.*

## 8 Temperley–Lieb closure of the boundary algebra

The boundary inherits not only a gauge algebra but a categorical structure: at the lock-in holonomy the relevant anyonic category truncates to a finite Temperley–Lieb algebra. This section fixes the deformation parameter from the lock-in mode and then establishes the three structural theorems — the graded space, the braid representation, and the invariance of the residue — that the gauge analysis relies on.

### 8.1 The deformation parameter

The lock-in mode  $n_* = 5$  carries holonomy  $\theta_* = \pi/n_*$  under parallel transport around  $S_\chi^1$ . The Majorana condition requires rational closure: the minimal closure angle must be a rational multiple of  $\pi$  (otherwise the bilinear  $\bar{\Xi}\Gamma\Xi$  would not be periodic in  $\tau$ ). The rank-five constraint forces the denominator of that rational to equal  $n_*$ . Together they determine the deformation parameter:

$$q = e^{i\pi/n_*}. \quad (49)$$

**Proposition 8.1** (Boundary admits a rank-five Fibonacci/TL representation). *The observable boundary of the rank-five Majorana projection admits a representation by the Temperley–Lieb algebra  $\text{TL}_4(\varphi)$  [12, 13], with deformation parameter  $q = e^{i\pi/n_*}$ . We construct the representation  $(F, R, U_i)$  below; we do not claim that this is the unique boundary exchange algebra, only that a compatible Fibonacci/TL representation exists at rank five.*

*Proof.* **Step 1: Truncation.** At  $q = e^{i\pi/n_*}$ :

$$[n_*]_q = \frac{\sin(n_* \cdot \pi/n_*)}{\sin(\pi/n_*)} = \frac{\sin \pi}{\sin(\pi/n_*)} = 0.$$

The Jones–Wenzl projector  $P_{n-1}$  satisfies  $P_{n-1} = 0$  when  $[n]_q = 0$  [13]. Here  $n = n_* = 5$ , so  $P_4 = 0$  and the Temperley–Lieb algebra truncates to  $\text{TL}_4$ .

**Step 2: Fibonacci fusion.** At  $q = e^{i\pi/5}$ , the quantum dimension of the simple object  $\tau$  is  $d_\tau = [2]_q = 2 \cos(\pi/5) = \varphi$  (the golden ratio). In the chosen Fibonacci realisation the fusion rule is  $\tau \otimes \tau = \mathbf{1} \oplus \tau$ , and the Verlinde formula gives fusion multiplicities  $\dim \text{Hom}(\tau^{\otimes n}, \mathbf{1}) = F_{n-1}$  [15], so the empty-channel multiplicity at  $n$  anyons is  $F_{n-1}$ .

**Step 3: Matching.** The two relevant dimensions both arise from the four-anyon fusion space. Using  $\tau \otimes \tau = \mathbf{1} \oplus \tau$  and the recurrence  $\tau^{\otimes n} = F_{n-1} \mathbf{1} \oplus F_n \tau$ , the four-anyon space splits as

$$\mathcal{H}_{\text{fus}}^{(4)} = V_{\mathbf{1}}^{\tau^4} \oplus V_{\tau}^{\tau^4}, \quad \dim V_{\mathbf{1}}^{\tau^4} = F_3 = 2, \quad \dim V_{\tau}^{\tau^4} = F_4 = 3, \quad (50)$$

so  $\dim \mathcal{H}_{\text{fus}}^{(4)} = 2 + 3 = 5$  reproduces the rank. The vacuum block  $V_{\mathbf{1}}^{\tau^4} \cong \mathbb{C}^2$  is the mixing plane  $\mathcal{H}_{\text{dyn}}$  on which  $\Theta_{\partial}$  is defined; the charged block  $V_{\tau}^{\tau^4} \cong \mathbb{C}^3$  carries the adjoint-sector dimension. (The earlier reading via  $\dim \text{Hom}(\tau^{\otimes 6}, \mathbf{1})$  is not used: the correct empty-channel multiplicity for  $N$  anyons is  $F_{N-1}$ , so six anyons give  $F_5 = 5$ , not 3; the value 3 is the charged multiplicity at four anyons, as above.)  $\square$   $\square$

## 8.2 Theorems G1–G3

**Theorem 8.2** (Gauge theorems G1–G3). *Under  $\text{TL}_4(\varphi)$  with the Majorana condition  $\Xi = C\bar{\Xi}$ , the following hold.*

**G1** (Graded space, by definition). *We define the graded boundary space as the direct sum of the two-anyon and four-anyon vacuum-channel spaces,*

$$\mathcal{H}_{\text{gr}} := \underbrace{V_{\mathbf{1}}^{\tau^2}}_{\mathcal{H}_{\text{proj}}, \dim=F_1=1} \oplus \underbrace{V_{\mathbf{1}}^{\tau^4}}_{\mathcal{H}_{\text{dyn}}, \dim=F_3=2}. \quad (51)$$

These are spaces of different anyon number, not canonical subspaces of a single fusion space, so (51) is a definition of  $\mathcal{H}_{\text{gr}}$ , not a decomposition forced by semisimplicity. The grading records the projection degree ( $n = 2$  for the projective readout,  $n = 4$  for the mixing plane);  $\mathcal{H}_{\text{proj}}$  carries the residue channel and  $\mathcal{H}_{\text{dyn}}$  the braid-dynamical plane.

The two-dimensional space  $\mathcal{H}_{\text{dyn}}$  is the mixing plane, distinct from the three-dimensional adjoint sector  $\mathfrak{su}(2)_{\xi}$ : the former is the braid-dynamical plane on which the channel count is read, the latter the adjoint algebra whose dimension is the numerator of  $\Theta_{\partial}$ . The adjoint Lie structure is supplied by the chiral doublet (Section 7).

**G2** (Fibonacci braid representation on  $\mathcal{H}_{\text{dyn}}$ ). *The four-anyon space  $\mathcal{H}_{\text{dyn}} = V_{\mathbf{1}}^{\tau^4}$  carries an action of  $B_4$  (generators  $\sigma_1, \sigma_2, \sigma_3$ ). We work with the restriction to the subgroup  $B_3 = \langle \sigma_1, \sigma_2 \rangle$  acting on the three active strands that span the two-dimensional vacuum channel; this restriction is all the mixing-plane analysis uses, and  $\sigma_3$  is not needed below. On this two-dimensional space the braid generators are built from the Fibonacci  $F$ - and  $R$ -matrices [16],*

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad F^{-1} = F, \quad R = \begin{pmatrix} e^{-4\pi i/5} & 0 \\ 0 & e^{3\pi i/5} \end{pmatrix}, \quad (52)$$

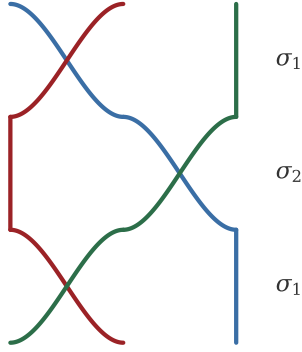
as

$$\rho(\sigma_1) = R, \quad \rho(\sigma_2) = FRF. \quad (53)$$

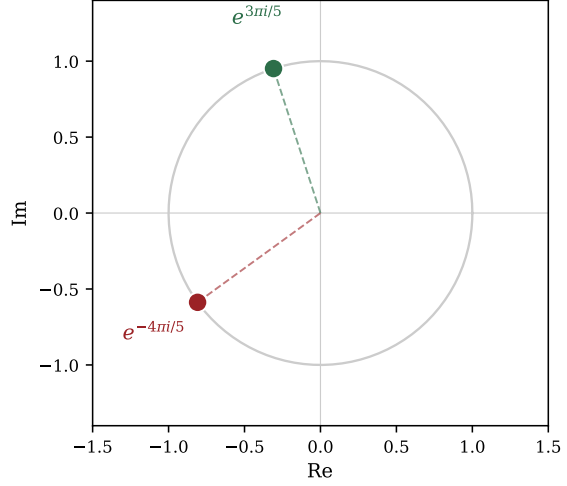
These satisfy the braid relation  $\rho(\sigma_1)\rho(\sigma_2)\rho(\sigma_1) = \rho(\sigma_2)\rho(\sigma_1)\rho(\sigma_2)$  and the quadratic relation  $(\rho(\sigma_i) - e^{-4\pi i/5})(\rho(\sigma_i) - e^{3\pi i/5}) = 0$ . The real matrix  $F$  is a change of basis, not a braid generator; the generators carry the Fibonacci phases through  $R$ . The Temperley–Lieb structure is recovered from the projectors  $e_i = (\rho(\sigma_i) - e^{3\pi i/5})/(e^{-4\pi i/5} - e^{3\pi i/5})$  and  $U_i = \varphi e_i$ , which satisfy  $U_i^2 = \varphi U_i$  and  $U_i U_{i\pm 1} U_i = U_i$ .

**G3** (Invariance of the residue as a scalar). *Let  $\rho^{(4)} = \rho_{\mathbf{1}} \oplus \rho_{\tau}$  be the full four-strand braid representation on  $V^{\tau^4} = V_{\mathbf{1}}^{\tau^4} \oplus V_{\tau}^{\tau^4}$  ( $\dim = 2 + 3 = 5$ ), of which  $\rho_{\mathbf{1}}$  is the two-dimensional irreducible block used in G2 and Proposition 9.3. The braiding preserves the total-charge sectors: writing  $P_{\mathbf{1}}$  and  $P_{\tau}$  for the orthogonal projectors onto the vacuum and  $\tau$  blocks, Schur’s lemma gives*

$$[\rho^{(4)}(b), P_{\mathbf{1}}] = [\rho^{(4)}(b), P_{\tau}] = 0, \quad P_{\mathbf{1}} \rho^{(4)}(b) P_{\tau} = P_{\tau} \rho^{(4)}(b) P_{\mathbf{1}} = 0, \quad (54)$$

(a) Fibonacci braid generators on  $B_3$ 

(b) Braiding eigenvalues on the unit circle



**Figure 1:** Left: the braid generators  $\sigma_1, \sigma_2$  on the mixing plane, realising the braid relation  $\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$  in the Fibonacci representation  $\rho(\sigma_1) = R$ ,  $\rho(\sigma_2) = FRF$ . Right: each generator satisfies the quadratic relation with eigenvalues  $e^{-4\pi i/5}$  and  $e^{3\pi i/5}$ , the Fibonacci braiding phases at  $q = e^{i\pi/n_*}$ .

for every braid  $b$ , where  $P_{\text{red}} = P_1$  is a total-charge projector (so  $[P_{\text{red}}, \rho^{(4)}(b)] = 0$  holds by construction). For every orthogonal projector  $P_{\text{red}}$  commuting with the braid representation, the complementary norm  $\sigma_{\text{cat}}(v) = \|(I - P_{\text{red}})v\|^2$  is braid-invariant:  $\sigma_{\text{cat}}(\rho(b)v) = \sigma_{\text{cat}}(v)$  by unitarity. This  $\sigma_{\text{cat}}$  is a braid-invariant categorical quadratic functional on  $V^{\tau^4}$  — a function of  $v$ , not a single number equal to 137; it is not a vector fixed by the braid (the two-dimensional Fibonacci representation has no common fixed line), and it is a separate object from the spectral residue  $\sigma_{\text{spec}} = \lfloor \alpha^{-1} \rfloor = 137$  of Section 5 — no intertwiner identifying the two is constructed here.

*Proof.* **G1.** G1 is a definition of the graded carrier  $\mathcal{H}_{\text{gr}}$  used in this paper, not a decomposition forced by semisimplicity: its two summands have different anyon number and are not canonical subspaces of one fusion space. Semisimplicity of the Fibonacci category determines only the *dimensions* of the two components,  $\dim V_1^{\tau^2} = F_1 = 1$  and  $\dim V_1^{\tau^4} = F_3 = 2$ , via the Verlinde formula.

**G2.** With  $\rho(\sigma_1) = R$  diagonal, the quadratic relation  $(\rho(\sigma_1) - e^{-4\pi i/5})(\rho(\sigma_1) - e^{3\pi i/5}) = 0$  is immediate; the same relation for  $\rho(\sigma_2) = FRF$  follows from  $F^{-1} = F$  and conjugation invariance of the spectrum. The braid relation  $\rho(\sigma_1)\rho(\sigma_2)\rho(\sigma_1) = \rho(\sigma_2)\rho(\sigma_1)\rho(\sigma_2)$  is verified by direct matrix multiplication using (52). The Temperley–Lieb generators  $U_i = \varphi e_i$  defined above satisfy  $U_i^2 = \varphi U_i$  and  $U_i U_{i\pm 1} U_i = U_i$ , again by direct computation, so the categorical closure is TL at loop value  $\varphi$  as claimed.

**G3.** The full braid representation  $\rho^{(4)}$  is block-diagonal in the total-charge projectors  $P_1, P_\tau$ , since braiding preserves the total anyonic charge of the strands; hence  $[\rho^{(4)}(b), P_1] = [\rho^{(4)}(b), P_\tau] = 0$  and the off-diagonal blocks vanish for every braid  $b$ . Take  $P_{\text{red}} = P_1$ , a sum of total-charge projectors, so  $[P_{\text{red}}, \rho^{(4)}(b)] = 0$ . Then unitarity of  $\rho^{(4)}(b)$  gives  $\sigma_{\text{cat}}(\rho^{(4)}(b)v) = \|(I - P_{\text{red}})\rho^{(4)}(b)v\|^2 = \|\rho^{(4)}(b)(I - P_{\text{red}})v\|^2 = \sigma_{\text{cat}}(v)$ . The scalar  $\sigma_{\text{cat}}$  is thus braid-invariant. It is not a braid-fixed vector: the non-Abelian two-dimensional representation has no common fixed line.  $\square$

The Temperley–Lieb structure G1–G3 protects the total-charge sectors and the scalar residue; it does *not* by itself derive the dimension ratio. The value  $\Theta_\partial = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|)$  is fixed in Section 6.2 from the rank and the channel count, independently of the categorical protection established here.

## 9 Fibonacci braid realisation of the observable channel count

Section 6.2 fixes  $\Theta_\partial = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|)$  once the number of observable adjoint channels is known. That number,  $|\mathcal{C}_{\text{obs}}| = \dim \mathfrak{su}(2) = 3$ , is obtained from the Majorana symplectic doublet in Section 2.3 and independently reproduced here by the Fibonacci braid representation under the categorical hypotheses (H1)–(H3) stated below. The braid realisation does not use the anisotropy of the lock-in tensor: it is purely a statement about the two-dimensional Fibonacci braid representation on the vacuum block  $\mathcal{H}_{\text{dyn}} = V_1^{\tau^4} \cong \mathbb{C}^2$ .

### 9.1 The Fibonacci sector as a conditional selection

Section 8 establishes that the rank-five lock-in holonomy  $q = e^{i\pi/n_*}$  with  $[n_*]_q = 0$  truncates Temperley–Lieb to  $\text{TL}_4(\varphi)$ , and that the boundary *admits* the Fibonacci braid representation. It does not establish that the observable exchange category must coincide with the Fibonacci subcategory: rank five and  $\varphi$  alone do not force this. The selection becomes a theorem under three categorical hypotheses, stated here as premises.

**Proposition 9.1** (Conditional selection of the Fibonacci sector). *Assume the observable exchange sector is*

(H1) *the unitary, semisimple quotient of  $\text{TL}(q)$  at  $q = e^{i\pi/5}$  (the level-three Temperley–Lieb–Jones quotient);*

(H2) *the even, self-dual, minimal subcategory generated by a single non-trivial simple object  $\tau$ ;*

(H3) *closed on the two channels  $\text{Irr} = \{\mathbf{1}, \tau\}$ .*

*Then the quantum dimension  $d_\tau = \varphi$  satisfies  $d_\tau^2 = 1 + d_\tau$  and forces the fusion rule  $\tau \otimes \tau = \mathbf{1} \oplus \tau$ ; the resulting category is Fibonacci, unique up to the two conjugate braiding chiralities [16], and its representation on  $V_1^{\tau^4} \cong \mathbb{C}^2$  generates  $M_2(\mathbb{C})$ .*

*Proof.* By (H3) the fusion of  $\tau$  with itself decomposes over  $\{\mathbf{1}, \tau\}$ :  $\tau \otimes \tau = N_{\tau\tau}^{\mathbf{1}} \mathbf{1} \oplus N_{\tau\tau}^\tau \tau$ . Self-duality  $\tau \cong \tau^*$  together with simplicity gives  $N_{\tau\tau}^{\mathbf{1}} = \dim \text{Hom}(\tau \otimes \tau, \mathbf{1}) = \dim \text{Hom}(\tau, \tau^*) = 1$ . Taking quantum dimensions (multiplicative under  $\otimes$ , additive under  $\oplus$ ),

$$d_\tau^2 = N_{\tau\tau}^{\mathbf{1}} + N_{\tau\tau}^\tau d_\tau = 1 + N_{\tau\tau}^\tau d_\tau.$$

With  $d_\tau = \varphi$  and  $\varphi^2 = 1 + \varphi$  this forces  $N_{\tau\tau}^\tau = 1$ , hence  $\tau \otimes \tau = \mathbf{1} \oplus \tau$ . This is the Fibonacci fusion ring; by (H1) the braided unitary structure on it is, up to the two complex-conjugate chiralities, the Fibonacci one [16]. Both chiralities give  $\rho(\sigma_1), \rho(\sigma_2)$  generating  $M_2(\mathbb{C})$  on  $V_1^{\tau^4}$  (Proposition 9.3).  $\square$

The two chiralities give the same envelope  $M_2(\mathbb{C})$  and the same count. What remains open is the selection of the hypotheses (H1)–(H3) from the projection operator  $\Pi$  itself: rank five and  $\varphi$  do not by themselves single out the minimal unitary even quotient, and we do not derive (H1)–(H3) from  $\Pi$  in this paper. The Fibonacci braid realisation of the count below is therefore conditional on (H1)–(H3); the count  $|\mathcal{C}_{\text{obs}}| = 3$  itself is primary from the symplectic doublet of Section 2.3 and does not depend on this selection.

### 9.2 Non-commutativity of the braid generators

On the mixing plane  $\mathcal{H}_{\text{dyn}} \cong \mathbb{C}^2$  the two elementary braids of Section 8 act by

$$\rho(\sigma_1) = R = \begin{pmatrix} r_1 & 0 \\ 0 & r_2 \end{pmatrix}, \quad \rho(\sigma_2) = F R F, \quad (55)$$

with eigenphases  $r_1 = e^{-4\pi i/5}$ ,  $r_2 = e^{3\pi i/5}$  and the involutive Fibonacci  $F$ -matrix

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad F^2 = \mathbb{1}, \quad (56)$$



where  $F^2 = \mathbb{K}$  follows from  $\varphi^{-2} + \varphi^{-1} = 1$ . Writing  $a = \varphi^{-1}$ ,  $b = \varphi^{-1/2}$  and evaluating  $FRF$  directly,

$$FRF = \begin{pmatrix} a^2 r_1 + b^2 r_2 & ab(r_1 - r_2) \\ ab(r_1 - r_2) & b^2 r_1 + a^2 r_2 \end{pmatrix}, \quad (FRF)_{12} = \varphi^{-3/2}(r_1 - r_2). \quad (57)$$

Since  $r_1 \neq r_2$ , the off-diagonal entry is non-zero, hence

$$[\rho(\sigma_1), \rho(\sigma_2)] = [R, FRF] \neq 0. \quad (58)$$

The braid representation on  $\mathcal{H}_{\text{dyn}}$  is therefore non-Abelian as a matter of computation, not assumption. This locates the non-Abelian structure in the braid topology, not in any anisotropy of the carrier; the role of the carrier anisotropy is confined to the selection of the Cartan direction  $T_3$  within the resulting algebra, as in Section 2.3.

### 9.3 Irreducibility and the Burnside envelope

**Lemma 9.2** (Irreducibility). *The representation of the braid group generated by  $\{\rho(\sigma_1), \rho(\sigma_2)\}$  on  $\mathcal{H}_{\text{dyn}} \cong \mathbb{C}^2$  is irreducible.*

*Proof.* A common invariant subspace would be a common eigenline. Since  $R = \text{diag}(r_1, r_2)$  with  $r_1 \neq r_2$ , the only  $R$ -invariant lines are  $\mathbb{C}e_1$  and  $\mathbb{C}e_2$ . By (57),  $(FRF)_{12} = (FRF)_{21} = \varphi^{-3/2}(r_1 - r_2) \neq 0$ , so  $FRF e_1$  has a non-zero  $e_2$ -component and  $FRF e_2$  a non-zero  $e_1$ -component: neither coordinate line is  $FRF$ -invariant. There is no common invariant line, and no invariant subspace of intermediate dimension exists in  $\mathbb{C}^2$ . The representation is irreducible.  $\square$

**Proposition 9.3** (Burnside envelope). *The complex associative algebra generated by  $\{\rho(\sigma_1), \rho(\sigma_2)\}$  is the full matrix algebra,*

$$\text{Alg}_{\mathbb{C}}\{R, FRF\} = \text{End}(\mathcal{H}_{\text{dyn}}) \cong M_2(\mathbb{C}). \quad (59)$$

*Proof.* By Burnside's theorem, a subalgebra of  $M_n(\mathbb{C})$  acting irreducibly on  $\mathbb{C}^n$  is all of  $M_n(\mathbb{C})$ . Lemma 9.2 supplies irreducibility for  $n = 2$ . Concretely, the four matrices  $\{\mathbb{K}, R, FRF, RFRF\}$  are linearly independent over  $\mathbb{C}$  (their images in  $\mathbb{C}^4$  under vectorisation have rank four), so they span  $M_2(\mathbb{C})$ .  $\square$

### 9.4 The compact real form and the count three

**Theorem 9.4** (Compact Lie algebra of the Fibonacci braid envelope). *The compact real form of the Burnside envelope (59) is  $\mathfrak{su}(2)$ , of real dimension three; it provides a second, independent realisation of the adjoint channel count*

$$|\mathcal{C}_{\text{obs}}| = \dim_{\mathbb{R}} \mathfrak{su}(2) = 3 \quad (60)$$

*already obtained from the symplectic doublet in Section 2.3.*

*Proof.* The derived algebra of the Burnside envelope is  $[M_2(\mathbb{C}), M_2(\mathbb{C})] = \mathfrak{sl}_2(\mathbb{C})$ , the traceless matrices. The braid generators (55) are unitary (diagonal phases for  $R$ ;  $FRF$  unitary as a conjugate of a unitary by the unitary involution  $F$ ), so the  $*$ -envelope is closed under the adjoint, and its compact real form is

$$\mathfrak{k}_{\text{br}} := \{X \in M_2(\mathbb{C}) : X^\dagger = -X, \text{tr } X = 0\} \cong \mathfrak{su}(2), \quad \dim_{\mathbb{R}} \mathfrak{k}_{\text{br}} = 3. \quad (61)$$

This  $\mathfrak{k}_{\text{br}}$  is the compact real form of the  $*$ -envelope, defined canonically from the associative algebra  $M_2(\mathbb{C})$ ; it is abstractly isomorphic to the algebra obtained in Section 2.3 from the symplectic reduction of the projected doublet (every  $\mathfrak{su}(2)$  is), but the present route obtains the dimension 3 from the braid envelope alone.  $\square$

**Remark 9.1** (What is forced and what remains). *Three points are kept distinct. (i) The channel count  $|\mathcal{C}_{\text{obs}}| = 3$  is supplied primarily by the metric-degeneracy chain of Section 2.3: the projected doublet  $E_\chi$  carries a  $U(2)$  stabiliser, the Majorana form reduces it to  $USp(2) \cong SU(2)$ , and  $\dim \mathfrak{su}(2) = 3$ . (ii) The braid envelope of Proposition 9.3 furnishes a second, independent operatorial realisation of the same count: by Lemma 9.2 and Proposition 9.3 the envelope is  $M_2(\mathbb{C})$ , whose compact  $*$ -form  $\mathfrak{k}_{\text{br}}$  has dimension 3 (the non-commutativity (58) is necessary for irreducibility but does not by itself fix the dimension; Lemma 9.2 and Proposition 9.3 are required). (iii) The identity of  $\mathfrak{k}_{\text{br}}$  with the projected gauge algebra  $\mathfrak{su}(2)_\xi$  — that the braid generators are the gauge generators — requires an intertwiner  $\mathcal{U} : \mathcal{H}_{\text{dyn}} \rightarrow S_\chi$  with  $\mathcal{U} \mathfrak{k}_{\text{br}} \mathcal{U}^{-1} = \mathfrak{su}(2)_\xi$ . The two algebras are abstractly isomorphic, so such a  $\mathcal{U}$  exists; fixing it canonically on the named generators requires a second oriented observable beyond the Cartan, and is not carried out here. The count 3 thus does not depend on the intertwiner; only the identification of the braid sector with the projected gauge sector does.*

The earlier categorical statement is corrected accordingly: the abstract Temperley–Lieb category does not by itself determine the boundary gauge Lie algebra, but its irreducible two-dimensional unitary Fibonacci braid representation has full matrix envelope  $M_2(\mathbb{C})$  (Proposition 9.3), whose traceless commutator algebra is  $\mathfrak{sl}_2(\mathbb{C})$  and whose compact real form is  $\mathfrak{su}(2)$  (Theorem 9.4). The Majorana structure identifies the compact form; the anisotropy selects its Cartan axis.

## 9.5 Assembly of the internal dimension and the dimension ratio

The three ingredients of the internal dimension are now each fixed structurally, on the counting space  $\mathcal{B}_{\text{count}}$  of Definition 6.4:

$$\dim \mathcal{B}_{\text{count}} = 2r + |\mathcal{C}_{\text{obs}}| = 2 \cdot 5 + 3 = 13, \quad (62)$$

where  $r = 5$  is the singular-value rank (threshold  $w_4 > \alpha > w_5$ , Section 3), the factor 2 is the chiral doubling of the carrier, and  $|\mathcal{C}_{\text{obs}}| = 3 = \dim \mathfrak{su}(2)_\xi$  is the adjoint count from the symplectic doublet (Section 2.3), independently reproduced under (H1)–(H3) by the Fibonacci braid envelope (Theorem 9.4). The dimension ratio of Section 6.2 then reads

$$\Theta_\partial = \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|} = \frac{3}{13}, \quad (63)$$

a form that uses only the rank, the chiral doubling, and the forced channel count. This is the derivation of  $\Theta_\partial$  carried by the paper.

**Remark 9.2** (Mathematical closure and the remaining physical step). *With (63) the internal value  $\Theta_\partial = 3/13$  is fixed by forced data:  $r$  and the chiral doubling from the rank-five structure, and  $|\mathcal{C}_{\text{obs}}| = 3$  from the symplectic doublet ( $USp(2) \cong SU(2)$ ), reproduced by the braid envelope. What remains open is not arithmetic but two structural identifications: the intertwiner  $\mathcal{U}$  of Remark 9.1, identifying the braid algebra with the projected gauge algebra; and the physical matching  $\Theta_\partial = \sin^2 \theta_W$ , which requires the kinetic conditions M1–M3 of Section 7 and is empirical in character. The present sections close the arithmetic of the numerator and denominator; the gauge-sector identity and the physical interpretation are deferred.*

## 10 Structural boundary nodes

The admissible modes occupy a bounded angular slab whose two endpoints act as structural nodes. This section locates the slab and records the mechanisms that keep the boundary classes separated, without invoking any boundary field equation.

## 10.1 The admissible slab and its boundary nodes

The boundary carries two structural nodes, at the covering mode and at the saturating mode: the angular positions  $\theta_{\text{cov}} = \pi/n_{\text{cov}}$  and  $\theta_{\text{sat}} = \pi/n_{\text{sat}}$  bound the admissible slab from above and below. These are structural endpoints of the admissible range, not dynamical domain walls: since no boundary field equation is used in this paper, we make no claim about a mass profile, a sign reversal, a penetration depth, or a localised zero mode. For the present purpose we use only the structural fact that the slab is bounded by these two nodes.

The admissible slab is the region  $\theta \in [\pi/n_{\text{sat}}, \pi/n_{\text{cov}}]$  (equivalently  $t \in [t_{11}^-, 0]$  in the Cayley coordinate of Section A), where all five modes coexist and the boundary bilinears are well-defined; the lock-in  $n_* = 5$  sits inside it. This is the fundamental half-slab of the full reflected structure developed in Section A; here we use only that it is bounded by two structural nodes.

## 10.2 Three protection mechanisms

What protects the boundary classes is not one mechanism but three, acting independently. (A fourth, Jackiw–Rebbi protection at the two structural nodes, is a downstream possibility but uses zero-mode dynamics not invoked here, so it is not counted.) The truncation  $[n_*]_q = 0$  sets the Jones–Wenzl projector  $P_4 = 0$ , so no physical state can leak out of the braid-graded sectors  $\mathcal{H}_{\text{proj}}, \mathcal{H}_{\text{dyn}}$  into higher Fibonacci sectors. The categorical scalar  $\sigma_{\text{cat}}(v) = \|(I - P_{\text{red}})v\|^2$  is braid-invariant (Theorem 8.2, G3), so no admissible braid transformation carries the complementary categorical sector into the  $P_{\text{red}}$  sector.

*Filter selection* by the zeros of  $P_4(z)$  in the causal filter  $H(z) = P_4(z)/(1 - \alpha z)$  [21]: the four zeros of the rank-five numerator polynomial select exactly the four physically distinct orbit classes, and the  $\mathbb{Z}_{10}$  monodromy of  $P_4$  provides an additional discrete protection of the orbit structure.

The confined class ( $n_{\text{conf}} = 7$  sector) carries zero boundary metric weight:  $\langle \psi_7, K_{\Pi} \psi_7 \rangle = 0$ . This is the metric signature of the confined kernel seed at the boundary level. The detailed spinorial lift of the confined class into  $\ker \Pi$  lies outside the present scope.

## 11 Two compatible descriptions

Beyond the symplectic derivation of the gauge algebra, the boundary structure admits two further descriptions of the invariant Abelian residue — a categorical truncation and a spectral filter — that share no common machinery. They are *compatible*, not proven equivalent. The single irreducible residue is the integer part of the filter pole,

$$\sigma_{\text{spec}} = \lfloor \alpha^{-1} \rfloor = 137, \quad \alpha^{-1} = \Lambda^2 = (F_5 + F_4\sqrt{5})^2 \quad (\text{Section 5}), \quad (64)$$

and the categorical side supplies, separately, a braid-invariant scalar  $\sigma_{\text{cat}}(v) = \|(I - P_{\text{red}})v\|^2$ . The two are structurally compatible but no intertwiner or common normalisation proving  $\sigma_{\text{cat}} = \sigma_{\text{spec}}$  is constructed here: the projector  $P_{\text{red}} = P_1$  of Theorem G3 is defined, but no normalisation is fixed in which  $\sigma_{\text{cat}}$  evaluates to 137. The internal dimension ratio  $\Theta_{\partial}$  is fixed in Section 6.2 by the channel count and the rank,  $\Theta_{\partial} = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|)$ ; the descriptions below concern the Abelian residue, not the ratio.

**Categorical description.** The lock-in holonomy  $q = e^{i\pi/n_*}$  truncates the Temperley–Lieb algebra to  $\text{TL}_4(\varphi)$ . Theorems G1–G3 give the graded space, the Fibonacci braid representation, and the braid invariance of the categorical scalar  $\sigma_{\text{cat}}$ . The categorical description supplies the channel count  $|\mathcal{C}_{\text{obs}}| = 3$  entering  $\Theta_{\partial}$  (Theorem 9.4); the scalar  $\sigma_{\text{cat}}$  is a separate object, and no map identifying it with  $\sigma_{\text{spec}}$  is constructed.

**Spectral description.** The causal filter

$$H(z) = \frac{P_4(z)}{1 - \alpha z}, \quad P_4(z) = \sum_{k=0}^4 \sigma_k z^k, \quad \sigma_k \propto \varphi^{-k}, \quad (65)$$

is holomorphic on  $|z| \leq 1$  (the pole is at  $z_\infty = \alpha^{-1}$ , far outside the disc). The pole location  $\alpha^{-1}$  is the continuous spectral threshold; its integer part  $\sigma_{\text{spec}} = \lfloor \alpha^{-1} \rfloor = 137$  is the spectral boundary residue. The four zeros of  $P_4$  encode the four boundary orbit classes. The  $\mathbb{Z}_{10}$  monodromy of  $P_4$  is the autosimilar clock of the projection [21].

The two descriptions are compatible on the invariant Abelian channel, the three adjoint forms (whose Lie structure is supplied by the chiral doublet, Section 7), and the confined kernel seed. Categorically  $\sigma_{\text{cat}}$  is a braid-invariant scalar; spectrally  $\sigma_{\text{spec}}$  is the integer residue of the boundary filter. The two constructions are structurally compatible, but no intertwiner or common normalisation proving their equality is constructed in this paper.

## 12 Conclusions

Given the finite-rank Majorana projection structure established in the companion papers, this paper constructs an observable boundary stabiliser algebra from a rank-five non-invertible projection of a Majorana bulk carrier onto a finite boundary, and an internal dimension ratio  $\Theta_\partial = 3/13$ . Its identification with the Standard-Model gauge sector is partial and conditional, as detailed below.

**State of the derivation.** The internal value  $\Theta_\partial = 3/13$  admits a derivation whose three ingredients are each fixed structurally: the rank  $r = 5$  by the singular-value threshold  $w_4 > \alpha > w_5$ ; the chiral doubling factor 2 by the Majorana double sheet; and the observable channel count  $|\mathcal{C}_{\text{obs}}| = 3$  by the Majorana symplectic doublet ( $USp(2) \cong SU(2)$ , Section 2.3), independently reproduced by the compact form of the Fibonacci braid envelope under (H1)–(H3) (Theorem 9.4), whence  $\Theta_\partial = |\mathcal{C}_{\text{obs}}|/(2r + |\mathcal{C}_{\text{obs}}|) = 3/13$  (equation (63)). The saturation mode  $n = 11$  and the generator norm 317 are fixed by the prime-lattice saturation  $\pi(11) = 5 = r$  grounded in the fixed point of Theorem 4.1, and the fine-structure scale is the pole of the boundary filter,  $\alpha^{-1} = \Lambda^2 = (F_5 + F_4\sqrt{5})^2 \approx 137.08$ , fixed by the Pincherle dilation operator (equation (32)).

Three items remain open, none arithmetic. (i) The *selection of the exchange sector*: the hypotheses (H1)–(H3) that single out the Fibonacci subcategory are not derived from the projection operator  $\Pi$  itself (Proposition 9.1); rank five and  $\varphi$  admit, but do not force, that sector. (ii) The *intertwiner*  $\mathcal{U} : \mathcal{H}_{\text{dyn}} \rightarrow S_\chi$  identifying the braid  $\mathfrak{su}(2)$  with the projected gauge  $\mathfrak{su}(2)_\xi$  (Remark 9.1): the braid envelope reproduces the channel count fixed by the symplectic doublet, but the identity of the braid sector with the projected gauge sector is not constructed. (iii) The *physical matching*  $\Theta_\partial = \sin^2 \theta_W$ , requiring the kinetic conditions M1–M3 of Section 7, which are empirical in character. Separately, a uniqueness theorem for the projection scale  $\Lambda$ , beyond its construction from the Pincherle increment and the Fibonacci curvature, is not claimed; this is a limit of the Pincherle construction, not a gauge gate.

The argument runs in three layers, each constraining the next. The self-conjugacy  $\Xi = C\bar{\Xi}$  forbids a complex phase rotation ( $e^{2i\theta} = 1$  on a Majorana spinor), so the only Abelian freedom is the real rotation  $U(1)_\eta$  of the chiral plane, and a conserved structure surviving projection of that phase requires a non-Abelian component (Section 2.3): the reality condition restricts the continuous symmetry content to a single non-Abelian factor  $SU(2)_\xi$ , with the real rotation  $U(1)_\eta$  as its Cartan. The rank-five projection fixes which modes reach the boundary, each admissible mode being the one meeting its closure condition — cover ( $n_{\text{cov}} = 2$ ), adjoint seed ( $n_{\text{adj}} = 3$ ), lock-in ( $n_* = 5$ ), confined kernel seed ( $n_{\text{conf}} = 7$ ), saturation ( $n_{\text{sat}} = 11$ ). The lock-in holonomy  $q = e^{i\pi/n_*}$  is the value at which the rank-five truncation closes, and is the holonomy at which

Temperley–Lieb reduces to  $\text{TL}_4(\varphi)$ , carrying the Fibonacci braid whose envelope reproduces the adjoint count.

The non-Abelian component of the boundary stabiliser is  $\mathfrak{su}(2)$ : the degenerate singular doublet of the boundary metric carries a  $U(2)$  stabiliser, and the Majorana skew form  $\varepsilon$  reduces it to  $USp(2) \cong SU(2)$ , the reduction being fixed since the symplectic condition  $U^T \varepsilon U = \varepsilon$  selects a unique compact real form. Its three generators, with the rank and the chiral doubling, give the internal dimension ratio

$$\Theta_\partial = \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|} = \frac{3}{13},$$

the content of Definition 6.1 and Corollary 6.2. The number is the trace the structure leaves behind; the structure is the ratio of the adjoint dimension to the full internal dimension, and every term in it is fixed by the projection geometry, with no free parameter, no coupling constant, and no symmetry-breaking input.

The honesty of the result is in its boundaries. The cascade is conditional on three stated inputs — the imported rank-five admissibility class, the symplectic doublet, and the categorical hypotheses (H1)–(H3) — and is rigid once they are adopted: the carrier fixes the transport symmetries, the rank fixes the admissible modes, the lock-in fixes the categorical truncation, each step leaving the next no freedom. Where the structure stops determining — the selection of the exchange sector from the projection, the intertwiner identifying the braid algebra with the projected gauge algebra, the matching of the ratio to the measured mixing angle — the paper marks the step open. The rigidity is in the conditional cascade, not in any claim that the inputs themselves are forced from nothing.

The result speaks to two open problems at once. For closed, background-independent frameworks of Wheeler–DeWitt type, the paper gives the associated readout problem a definite answer at the level of gauge algebra: once the constrained carrier is accessed through a finite rank-five Majorana projection, the observable gauge-sector skeleton and the internal boundary dimension ratio are fixed once the imported admissibility class, the symplectic doublet, and the categorical hypotheses are adopted. This is not a proof that nature must realise a Wheeler–DeWitt bulk; it is a conditional result about what a finite boundary can read if the fundamental carrier is closed, constrained, and accessed only through a non-invertible projection. In that conditional sense it makes the programme of relational observables [3, 4, 10] concrete — the gauge algebra emerges from the boundary readout rather than being assumed. For the Standard Model, the construction yields a boundary gauge vocabulary — a non-Abelian adjoint channel  $SU(2)_\xi$  with its Abelian Cartan direction  $U(1)_\eta$ , and a confined kernel seed — together with the internal dimension ratio  $\Theta_\partial = 3/13$ , fixed by the rank, the chiral doubling, and the channel count. We do not claim the Standard-Model skeleton  $U(1)_Y \times SU(2)_L \times SU(3)_c$  as derived: the adjoint Lie algebra does close from the metric degeneracy of the boundary — the degenerate singular doublet carries  $SU(2)$  once the Majorana form reduces its  $U(2)$  stabiliser, and  $\text{End}_0(S_\chi) \cong \mathfrak{su}(2)$  realises it, with the projection anisotropy fixing the Cartan — but identifying these internal factors with the physical electroweak group requires an additional independent  $U(1)_Y$  sector — the real rotation  $U(1)_\eta$  being the Cartan of  $SU(2)_\xi$ , not a hypercharge factor — together with the identification  $SU(2)_\xi \simeq SU(2)_L$ , the left-chiral action, the hypercharge normalisation, the absence of kinetic mixing, and the common Hessian factor (Section 7). The internal structural result stands independently of all of these.

**Outside the present scope.** Physical masses of the electroweak gauge bosons and the Higgs boson require the electroweak vacuum expectation value and dimensional normalisation. Running of coupling constants, Yukawa matrices, CKM and PMNS mixing matrices, and full QCD dynamics are downstream. The physical fine-structure constant as a self-consistent fixed point, and the spinorial lift of the confined mode  $n_{\text{conf}} = 7$  into  $\ker \Pi$ , are left to separate papers in this series.

**Summary of status.** The result of this paper is, schematically,

$$\text{rank-five Majorana projection} \Rightarrow \text{observable stabiliser algebra} \Rightarrow \mathfrak{su}(2)_\xi \supset \mathfrak{u}(1)_\eta,$$

and

$$\Theta_\partial = \frac{|\mathcal{C}_{\text{obs}}|}{2r + |\mathcal{C}_{\text{obs}}|} = \frac{3}{13} \quad (\text{internal boundary dimension ratio}).$$

The individual claims carry distinct epistemic status:

- canonical denominator 13 ( $\dim \mathcal{B}_{\text{count}}$ ): *closed* (Definition 6.4, Proposition 6.3);
- identification  $\Theta_\partial = \sin^2 \theta_W$ : *open*, conditional on M1–M3 (Section 7);
- equality of the two residues  $\sigma_{\text{cat}}, \sigma_{\text{spec}}$ : *open*, no intertwiner constructed (Section 11).

The internal ratio is an internal theorem of the boundary algebra; it is not a tree-level electroweak mixing ratio, and the present paper does not claim it as one.

*The author acknowledges no external funding.*

## A Independent dynamical compatibility: the conformal slab and the kernel quadrature

The admissible modes were placed on the holonomy angle  $\theta$  in Section 10. That description is the fundamental domain of a reflection; the full structure, and the dynamical reading that goes with it, becomes transparent in the Cayley coordinate. This section is a separate dynamical layer: it invokes a boundary Dirac operator, and the algebraic results of the earlier sections neither assume nor depend on it.

### A.1 The Cayley coordinate and the Majorana reflection

Let  $\theta \in (0, \pi)$  be the holonomy coordinate on the compact internal fibre  $S_\chi^1$ . The chiral shell is  $S_\chi^2 \simeq \mathbb{CP}^1$ ; fixing the azimuthal angle  $\phi$  selects a meridian of this shell, and  $t$  below is the latitudinal coordinate along that meridian, with  $\phi$  a spectator. On the meridian set

$$z = e^{i\phi} \tan \frac{\theta}{2}, \quad t := \log |z| = \log \tan \frac{\theta}{2}, \quad \text{sech } t = \sin \theta, \quad \tanh t = -\cos \theta. \quad (66)$$

The Cayley line is thus an embedding  $\mathbb{R}_t \hookrightarrow \mathbb{CP}^1$  of a single meridian, not the whole shell. The physical Majorana reflection is the anti-holomorphic involution

$$\mathcal{R}(z) = \frac{1}{\bar{z}}, \quad \theta \mapsto \pi - \theta, \quad t \mapsto -t, \quad \phi \mapsto \phi, \quad (67)$$

since  $\log \tan((\pi - \theta)/2) = -\log \tan(\theta/2)$ . It is an *equatorial* reflection: it fixes the equator  $|z| = 1$  pointwise. It is not the antipodal map  $z \mapsto -1/\bar{z}$ , which fixes no point of the equator.

### A.2 Two projective charts

The two affine charts  $\mathcal{U}_0, \mathcal{U}_\infty$  of  $\mathbb{CP}^1$  are related on their overlap by the holomorphic coordinate transition

$$w = \frac{1}{z}. \quad (68)$$

This coordinate transition must be distinguished from the physical reflection  $\mathcal{R}(z) = 1/\bar{z}$  of (67). Both invert the radial coordinate  $|z| \mapsto |z|^{-1}$  and therefore induce  $t \mapsto -t$ , but they differ on the phase ( $w = 1/z$  sends  $\phi \mapsto -\phi$ , while  $\mathcal{R}$  keeps  $\phi$ ), and only  $\mathcal{R}$  is the Majorana reflection. The labels  $0, \infty$  denote projective charts, not holonomy windings: the winding law  $\theta_n = \pi/n$  is undefined at  $n = 0$  and gives  $\theta_1 = \pi$  (a conformal pole), neither an observable mode. The holonomy classes are  $n \in \{2, 3, 5, 7, 11\}$ .



### A.3 The Cayley–Hardy bulk completion

The primary boundary–bulk object is the constant-modulus completion

$$\Xi_{\text{bulk}}(t) = \varphi e^{-i \text{gd}(\Gamma_\tau t)}, \quad (69)$$

where  $\text{gd}$  is the Gudermannian function and  $\Gamma_\tau$  a fixed scale. Using the exact identities

$$\cos(\text{gd } x) = \text{sech } x, \quad \sin(\text{gd } x) = \tanh x, \quad (70)$$

the completion splits into its even and odd quadratures

$$\Xi_{\text{bulk}} = \Psi_{\text{bdy}} + \Xi_{\text{ker}}, \quad \Psi_{\text{bdy}}(t) = \varphi \text{sech}(\Gamma_\tau t), \quad \Xi_{\text{ker}}(t) = -i\varphi \tanh(\Gamma_\tau t). \quad (71)$$

These are the even and odd Cayley–Hardy quadratures of one analytic bulk signal —  $\Psi_{\text{bdy}}$  even and  $\Xi_{\text{ker}}$  odd under  $\mathcal{R}$ , since  $\text{sech}(-t) = \text{sech } t$  and  $\tanh(-t) = -\tanh t$ . They are *not* the two spinor components of a Jackiw–Rebbi zero mode (Section A.5 below). Their modulus is conserved,

$$|\Xi_{\text{bulk}}|^2 = |\Psi_{\text{bdy}}|^2 + |\Xi_{\text{ker}}|^2 = \varphi^2 (\text{sech}^2 + \tanh^2)(\Gamma_\tau t) = \varphi^2, \quad (72)$$

the hyperbolic Pythagorean identity: the boundary readout and the kernel quadrature are the two legs of a single conserved unit. At the poles ( $t \rightarrow \pm\infty$ ) the readout vanishes and the kernel saturates,  $\Xi_{\text{ker}} \rightarrow \mp i\varphi$ ; at the equator ( $t = 0$ ) the readout is maximal,  $\Psi_{\text{bdy}} = \varphi$ , and the kernel vanishes. The poles are kernel-dominant charts; the equator is readout-dominant. The relation (72) may be read as a bound/kernel partition, but  $\Xi_{\text{ker}}$  remains a complex amplitude, not merely a continuum density.

### A.4 The reflected slab

The reflection-fixed locus is the equator

$$\theta_2 = \frac{\pi}{2}, \quad t_2 = 0, \quad \mathcal{R}\left(\frac{\pi}{2}\right) = \frac{\pi}{2}, \quad (73)$$

where  $\Psi_{\text{bdy}}$  is maximal and  $\Xi_{\text{ker}} = 0$ . The covering mode  $n = 2$  is therefore the equatorial readout plane and the gluing locus of the two charts — not an endpoint of the slab. The saturation class  $n = 11$  has two reflected representatives of the *same* holonomy class,

$$\theta_{11}^- = \frac{\pi}{11}, \quad \theta_{11}^+ = \pi - \frac{\pi}{11} = \frac{10\pi}{11}, \quad t_{11}^+ = -t_{11}^-, \quad (74)$$

so the full reflected slab is

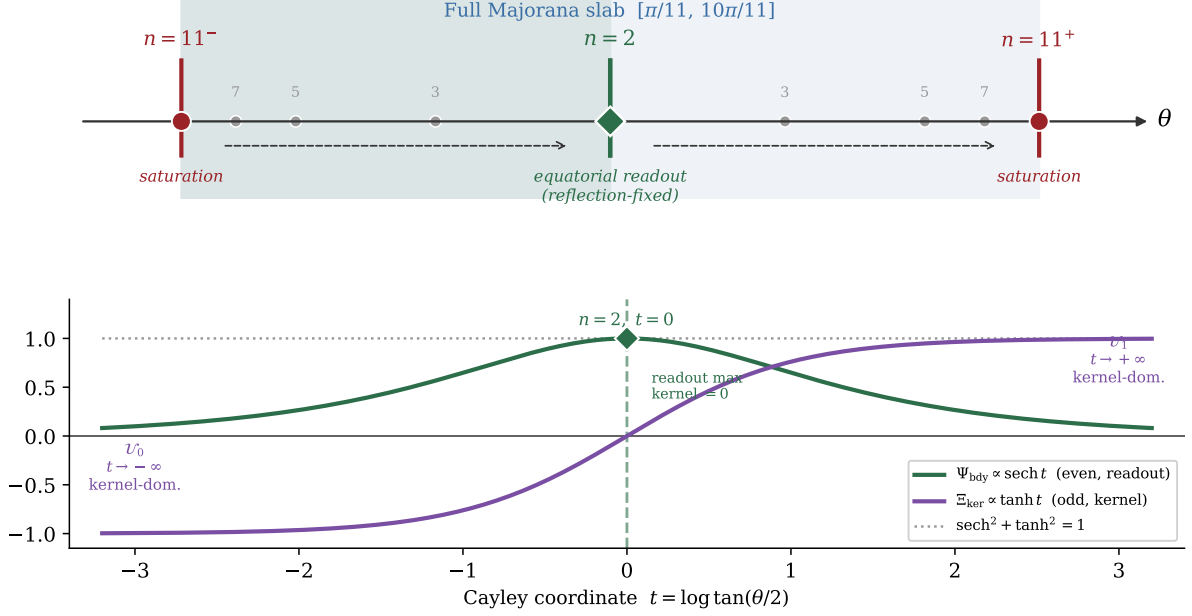
$$\mathcal{F}_{\text{full}} = \left[ \frac{\pi}{11}, \frac{10\pi}{11} \right] = \mathcal{F}_- \cup_{n=2} \mathcal{F}_+, \quad \mathcal{F}_- = \left[ \frac{\pi}{11}, \frac{\pi}{2} \right], \quad (75)$$

with the equator  $n = 2$  at the centre and  $n = 11^\pm$  as its reflected images,  $n = 11^- \rightarrow n = 2 \rightarrow n = 11^+$ . The region  $[\pi/11, \pi/2]$  used in Section 10 is the fundamental half-slab  $\mathcal{F}_-$ , the quotient under the  $\mathbb{Z}_2$  reflection; the two endpoints are not  $n = 11$  and a distinct mode, but the two reflected images of the single saturation class.

### A.5 Jackiw–Rebbi compatibility

The Cayley–Hardy decomposition above is kinematic. A *distinct* dynamical realisation follows from the one-dimensional Dirac operator

$$\mathcal{D} = -i\sigma_1 \partial_t + m(t)\sigma_3, \quad m(t) = m_0 \tanh(\Gamma_\tau t), \quad (76)$$



**Figure 2:** Top: the full reflected slab on the holonomy angle  $\theta \in [\pi/11, 10\pi/11]$ . The covering mode  $n = 2$  ( $\theta = \pi/2$ , green diamond) is the equatorial readout plane and the reflection-fixed centre; the saturation class  $n = 11$  appears as two reflected images  $n = 11^\pm$  (red), and the inner modes  $n = 3, 5, 7$  likewise occur in reflected pairs. The fundamental half-slab  $[\pi/11, \pi/2]$  of Section 10 is the left half. Bottom: the even readout quadrature  $\Psi_{\text{bdy}} \propto \text{sech}(\Gamma_\tau t)$  and the odd kernel quadrature  $\Xi_{\text{ker}} \propto \tanh(\Gamma_\tau t)$  on the Cayley coordinate, with the conserved modulus  $\text{sech}^2 + \tanh^2 = 1$ ; the conformal poles  $\mathcal{U}_0, \mathcal{U}_\infty$  ( $t \rightarrow \mp\infty$ ) are kernel-dominant.

whose mass changes sign at the equator. Multiplying  $\mathcal{D}\psi = 0$  by  $i\sigma_1$  gives  $\partial_t\psi + m(t)\sigma_2\psi = 0$ . With  $\chi_+$  the  $+1$  eigenspinor of  $\sigma_2$ ,

$$\sigma_2\chi_+ = \chi_+, \quad \chi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad (77)$$

the normalisable zero mode is

$$\psi_{\text{JR}}(t) = \mathcal{N} \exp \left[ - \int_0^t m(s) ds \right] \chi_+ = \mathcal{N} [\cosh(\Gamma_\tau t)]^{-m_0/\Gamma_\tau} \chi_+, \quad (78)$$

normalisable for  $m_0 > 0$ . For  $m_0 = \Gamma_\tau$  its envelope is exactly  $\text{sech}(\Gamma_\tau t)$ , the readout quadrature of (71), and  $\int_{-\infty}^{\infty} \text{sech}^2(\Gamma_\tau t) dt = 2/\Gamma_\tau$ . Both spinor components of  $\psi_{\text{JR}}$  are proportional to  $\text{sech}$ ; the odd function  $\tanh(\Gamma_\tau t)$  is the kink mass profile, *not* a second component of the zero mode. The QGT identification of the kernel quadrature with the kink background is a constitutive relation,

$$m(t) = \frac{i m_0}{\varphi} \Xi_{\text{ker}}(t), \quad (79)$$

not an automatic identity of Dirac theory: it is the statement that makes the invisible kernel quadrature act as the mass background of the wall. With this operator present,  $n = 2$  is a Jackiw–Rebbi domain wall and  $\text{sech}$  its localised zero mode; without it,  $n = 2$  is the geometric reflection-fixed equator of the previous subsections.

## A.6 Oriented boundary–kernel transfer

The reflection  $\mathcal{R} : t \mapsto -t$  leaves the readout invariant and the kernel modulus invariant, but reverses the oriented kernel quadrature. To make this quantitative without an unjustified Noether

current, define the normalised readout and kernel fractions

$$p_{\text{bdy}}(t) = \text{sech}^2(\Gamma_\tau t), \quad p_{\text{ker}}(t) = \tanh^2(\Gamma_\tau t), \quad p_{\text{bdy}} + p_{\text{ker}} = 1. \quad (80)$$

The oriented transfer rate is their flux,

$$J_{\text{bdy} \rightarrow \text{ker}}(t) := \partial_t p_{\text{ker}}(t) = 2\Gamma_\tau \tanh(\Gamma_\tau t) \text{sech}^2(\Gamma_\tau t), \quad J_{\text{bdy} \rightarrow \text{ker}}(-t) = -J_{\text{bdy} \rightarrow \text{ker}}(t), \quad (81)$$

odd under  $\mathcal{R}$  and positive on  $t > 0$  (transfer from readout to kernel), negative on the reflected branch  $t < 0$ . If an additional effective dynamics selects a causal branch, the reflected branch is excluded from that effective evolution, and the symmetry  $t \leftrightarrow -t$  becomes an oriented transfer. Non-invertibility of  $\Pi$  alone does not establish causal selection of a branch; an effective dynamics is required. We claim no more than this: a thermodynamic entropy-production inequality  $\dot{S}_{\text{phys}} \geq 0$  does not follow from the Cayley decomposition alone. Indeed the binary entropy  $S_2 = -p_{\text{bdy}} \log p_{\text{bdy}} - p_{\text{ker}} \log p_{\text{ker}}$  has an odd derivative that does not keep a fixed sign on  $t > 0$ ; a thermodynamic arrow would require an additional coarse-graining map or dissipative semigroup, which is outside the present scope.

## References

- [1] B. S. DeWitt, *Quantum Theory of Gravity. I. The Canonical Theory*, Phys. Rev. **160**, 1113 (1967).
- [2] J. A. Wheeler, *Superspace and the Nature of Quantum Geometrodynamics*, in *Battelle Rencontres*, eds. C. DeWitt and J. Wheeler (Benjamin, New York, 1968), pp. 242–307.
- [3] C. Rovelli, *Relational Quantum Mechanics*, Int. J. Theor. Phys. **35**, 1637 (1996). [arXiv:quant-ph/9609002](#).
- [4] E. Adlam and C. Rovelli, *Information is Physical: Cross-Perspective Links in Relational Quantum Mechanics*, Philos. Phys. **1**, 4 (2023). [arXiv:2203.13342](#).
- [5] R. B. Griffiths, *Consistent Quantum Theory* (Cambridge University Press, 2002); R. Omnès, *The Interpretation of Quantum Mechanics* (Princeton University Press, 1994).
- [6] M. Gell-Mann and J. B. Hartle, *Quantum Mechanics in the Light of Quantum Cosmology*, in *Complexity, Entropy, and the Physics of Information*, ed. W. H. Zurek (Addison-Wesley, 1990), pp. 425–458. [arXiv:1803.04605](#).
- [7] W. H. Zurek, *Decoherence, Einselection, and the Quantum Origins of the Classical*, Rev. Mod. Phys. **75**, 715 (2003). [arXiv:quant-ph/0105127](#).
- [8] W. H. Zurek, *Quantum Darwinism*, Nature Phys. **5**, 181 (2009).
- [9] C. J. Isham, *Canonical Quantum Gravity and the Problem of Time*, in *Integrable Systems, Quantum Groups, and Quantum Field Theories*, eds. L. A. Ibort and M. A. Rodríguez (Kluwer, 1993), pp. 157–287. [arXiv:gr-qc/9210011](#).
- [10] C. J. Isham and K. V. Kuchař, *Representations of Spacetime Diffeomorphisms. II. Canonical Geometrodynamics*, Ann. Phys. **164**, 316 (1985).
- [11] E. Anderson, *The Problem of Time in Quantum Gravity*, Ann. Phys. (Berlin) **524**, 757 (2012). [arXiv:1206.2403](#).
- [12] H. N. V. Temperley and E. H. Lieb, *Relations between the “Percolation” and “Colouring” Problem*, Proc. R. Soc. Lond. A **322**, 251 (1971).

- [13] V. F. R. Jones, *Index for Subfactors*, Invent. Math. **72**, 1 (1983).
- [14] M. Cipolla, La determinazione asintotica dell' $n^{\text{imo}}$  numero primo, *Rendiconto dell'Accademia delle Scienze Fisiche e Matematiche (Napoli)* **8** (1902) 132–166.
- [15] M. H. Freedman, A. Kitaev, and Z. Wang, *Simulation of Topological Field Theories by Quantum Computers*, Commun. Math. Phys. **227**, 587 (2002).
- [16] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, *Non-Abelian Anyons and Topological Quantum Computation*, Rev. Mod. Phys. **80**, 1083 (2008).
- [17] R. Jackiw and C. Rebbi, *Solitons with Fermion Number  $\frac{1}{2}$* , Phys. Rev. D **13**, 3398 (1976).
- [18] P. Camelia, *The Physics of an Incomplete Crossing: A Single Conservation Law under Restricted Access*, preprint (2026). <https://www.researchgate.net/publication/407084785>.
- [19] P. Camelia, *A Quantum Gauge Theory of Time (QGT): Foundations from a Noetherian Balance*, preprint (2026). <https://www.researchgate.net/publication/406309056>.
- [20] P. Camelia, *Uniqueness of the Rank-5 Projection in the Quantum Gauge Theory of Time*, preprint (2026). <https://www.researchgate.net/publication/407045507>.
- [21] P. Camelia, *Projection-Induced SVD and Boundary Metric: A Rank-Five Fibonacci-Autosimilar Characterization*, preprint (2026). <https://www.researchgate.net/publication/406312686>.
- [22] P. Camelia, *What a Finite Observer Cannot Know: Structural Constraints on Observable Algebras from Non-Invertible Projections*, preprint (2026). <https://www.researchgate.net/publication/407042683>.
- [23] P. Camelia, *The Fibonacci–Mellin Transform: Vandermonde Structure, Non-Isometry, and Finite-Field Orthogonality*, companion preprint (2026).